

# Calculus Done Taiwan

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## 0.1 Preface

*Done Taiwan Series* is a series of books that aims to help students prepare for the entrance examinations for top mathematics master's programs in Taiwan. The series currently consists of two books: *Linear Algebra Done Taiwan* and *Advanced Calculus Done Taiwan*. Each includes solutions and corrections to the exams administered by National Taiwan University over the past 10 years.

The one you are currently reading is *Advanced Calculus Done Taiwan*. The two books differ significantly in structure and style. In this book, there are two chapters. The first chapter develops some theory covered in the exams, but it is mostly written to be referenced by Chapter 2. Chapter 1 is therefore best used as reference material for Chapter 2 rather than read as a stand-alone chapter.

In this book, you will constantly see brown text like this:

[Click me to go to the last page of the book](#)

They are to make cross-references easier to follow. For example, you may read something like "By [Dirichlet's Test](#), we know  $\sum z_n(b_n - B)$  converge...", and clicking [Dirichlet's Test](#) would lead you to a statement and possibly a proof of [Dirichlet's Test](#).

Just like all mathematical texts, this book may contain some minor errors. If you believe you see an error or are confused by my writing, feel free to contact me. My email address is:

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# Chapter 1

## Selected topics

### 1.1 Compactness

Let  $X$  be a topological space. We say  $X$  is **compact** if every of its open cover has a finite subcover. We say  $X$  is **sequentially compact** if every sequence in  $X$  has a **convergent** subsequence. We say  $X$  is **limit point compact** if every infinite subset has a limit point.

Let  $M$  be a metric space. We say  $M$  is **totally bounded** if for all  $\epsilon$ , space  $M$  can be covered by finite number of open balls of radius  $\epsilon$ . We say  $M$  is **complete**, if every Cauchy sequence in  $M$  converge.

**Equivalent Definition 1.1.1. (Compactness of metric space)** Let  $M$  be a metric space. The followings are equivalent:

- (i)  $M$  is compact.
- (ii)  $M$  is limit point compact.
- (iii)  $M$  is sequentially compact.
- (iv)  $M$  is totally bounded and complete.

*Proof.* (i)  $\implies$  (ii): The argument by nature relies on contradiction. Assume for a contradiction that there exists some sequence  $(x_n)$  of distinct elements in  $M$  whose image has no limit point<sup>1</sup>. For each  $x \in M$ , since  $x$  is not a limit point of  $\{x_n\}$ , there exists some open ball centered at  $x$  that contains at most one point of  $\{x_n\}$ , namely  $x$  itself if and only if it appears in the sequence. The collection of these open balls is then a open cover of  $M$  that has no finite subcover of  $M$  because  $\{x_n\}$  is infinite, a contradiction.

---

<sup>1</sup>Note that finite metric spaces are trivially limit point compact.

(ii)  $\implies$  (iii) is clear.

(iii)  $\implies$  (iv): It is easy to show  $M$  is complete. We now show  $M$  is totally bounded. Assume  $M$  is not for a contradiction. This implies the existence of some  $\epsilon$  and some sequence of distinct elements in  $M$  whose every points are  $\epsilon$ -away. Such sequence has no convergent subsequence, a contradiction.

(iv)  $\implies$  (i): Again the argument by nature relies on contradiction. Assume for one that  $(U_\alpha)$  is an open cover of  $M$  that has no finite subcover. Due to totally boundedness of  $M$ , we know  $M$  can be covered by some finite set of radius 1 open balls. Since  $(U_\alpha)$  has no finite subcover, one of these radius 1 open balls can't be covered by finite numbers of  $(U_\alpha)$ . Denote the center of that radius 1 open ball by  $x_1$ . Clearly,  $B_1(x_1)$  is again totally bounded, so it can be covered by some finite set of radius  $\frac{1}{2}$  open balls centered inside. Again, because  $B_1(x_1)$  can not be covered by finite numbers of  $(U_\alpha)$ , one of these radius  $\frac{1}{2}$  open balls can't be covered by finite number of  $(U_\alpha)$ . Denote the center of that radius  $\frac{1}{2}$  open ball by  $x_2$ . Repeating the same procedure, we have a Cauchy sequence  $\{x_n\} \subseteq M$ . Let  $x$  be the limit of this Cauchy sequence. Fix some  $\alpha$  such that  $x \in B_\epsilon(x) \subseteq U_\alpha$ . Clearly, we may find  $N$  large enough, for example  $N > 2/\epsilon$ , so that  $B_{\frac{1}{N}}(x_N) \subseteq B_\epsilon(x) \subseteq U_\alpha$ , a contradiction to the construction of  $B_{\frac{1}{N}}(x_N)$ . ■

**Theorem 1.1.2. (Dini's theorem)** Let  $X$  be a compact topological space. If continuous  $f_n : X \rightarrow \mathbb{R}$  monotonically pointwise converge to a continuous  $f : X \rightarrow \mathbb{R}$ . Then the convergence is uniform.

*Proof.* Skipped. ■

## 1.2 Arzelà–Ascoli theorem

Let  $X$  be a topological space,  $Y$  a metric space, and  $\mathcal{F}$  a family of function from  $X$  to  $Y$ . We say  $\mathcal{F}$  is **pointwise equicontinuous** if for all  $\epsilon$  and  $x \in X$ , there exists a neighborhood  $U \ni x$  such that  $d(f(x), f(y)) \leq \epsilon$  for all  $y \in U$  for all  $f \in \mathcal{F}$ . Give  $X$  a compatible metric. We say  $\mathcal{F}$  is **uniform equicontinuous** if for all  $\epsilon$  there exists  $\delta$  such that  $d(f(x), f(y)) \leq \epsilon$  for all  $\delta$ -close  $x, y \in X$  and  $f \in \mathcal{F}$ . Clearly uniform equicontinuity implies pointwise equicontinuity and that every  $f \in \mathcal{F}$  is uniformly continuous. The converse isn't true.

**Example 1.2.1.** Let  $\varphi(t) \triangleq \max\{0, 1 - |t|\}$ . Define

$$f_n(x) \triangleq n \cdot \varphi(n(x - n)), \quad n \in \mathbb{N}$$

Every  $f_n$  is uniformly continuous on  $\mathbb{R}$  and the collection  $\mathcal{F} = \{f_n : \mathbb{R} \rightarrow \mathbb{R} | n \in \mathbb{N}\}$  is pointwise equicontinuous, but  $\mathcal{F}$  is not uniform equicontinuous.

**Theorem 1.2.2. (Pointwise equicontinuity implies uniform equicontinuity on compact domain)** Let  $X, Y$  be two metric spaces and  $\mathcal{F}$  a pointwise equicontinuous collection of functions from  $X$  to  $Y$ . If  $X$  is compact, then  $\mathcal{F}$  is moreover uniform equicontinuous.

*Proof.* Fix  $\epsilon$ . Because  $\mathcal{F}$  is pointwise equicontinuous and  $X$  is compact, we know there exists a finite open cover  $\mathcal{U} \triangleq \{B_{\delta_1/2}(x_1), \dots, B_{\delta_n/2}(x_n)\}$  of  $X$  such that for all fixed  $i$ , we have:

$$d(f(x_i), f(y)) \leq \frac{\epsilon}{2}, \quad \text{for all } y \in B_{\delta_i}(x_i) \text{ and } f \in \mathcal{F}$$

Let  $x, y \in X$  satisfies  $d(x, y) < \min \delta_i/2$ . Because  $\mathcal{U}$  forms an open cover, we know  $x \in B_{\delta_i/2}(x_i)$  for some  $i$ . Then clearly,  $y \in B_{\delta_i}(x_i)$ . This now give us:

$$d(f(x), f(y)) \leq d(f(x_i), f(x)) + d(f(x_i), f(y)) \leq \epsilon, \quad \text{for all } f \in \mathcal{F}$$

■

**Theorem 1.2.3. (Arzelà–Ascoli theorem)** Let  $X$  be a compact metric space, and let  $\mathcal{F}$  be a collection of continuous function from  $X$  to  $\mathbb{R}^n$ . If  $\mathcal{F}$  is pointwise equicontinuous and pointwise bounded, then:

- (I)  $\mathcal{F}$  is uniformly bounded.
- (II) Every sequence  $\{f_n\} \subseteq \mathcal{F}$  has a uniformly convergent subsequence.

or equivalently,  $\mathcal{F}$  is compact given the uniform norm topology.

*Proof.* Skipped.

■

**Theorem 1.2.4. (Sufficient conditions for uniform equicontinuity)** Let  $M$  be a metric space and  $\mathcal{F}$  a collection of function from  $M$  to  $\mathbb{R}$ . If for some  $K > 0$ , every  $f \in \mathcal{F}$  is  $K$ -Lipschitz, then  $\mathcal{F}$  is uniform equicontinuous. In particular, if  $M$  is an open subset of  $\mathbb{R}$  and  $\mathcal{F}$  is a collection of differentiable functions whose derivatives are uniformly bounded by some  $K$ :

$$|f'(x)| \leq K, \quad \text{for all } x \in U \text{ and } f \in \mathcal{F} \quad (1.1)$$

then  $f \in \mathcal{F}$  are all  $K$ -Lipschitz, thus making  $\mathcal{F}$  uniform equicontinuous.

*Proof.* If the functions are all  $K$ -Lipschitz, then  $\delta = \frac{\epsilon}{K}$  suffices to show uniform equicontinuity. If the functions are all differentiable and subject to **condition 1.1**, then **MVT** shows that they are all  $K$ -Lipschitz. ■

**Theorem 1.2.5. (Application scene of Arzelà–Ascoli theorem)** Let  $X, Y$  be two metric spaces, and  $\{f_n\}$  a sequence of functions from  $X$  to  $Y$ . If every subsequence contains a uniformly convergent subsequence, then every pointwise convergent subsequence uniformly converge.

*Proof.* Assume for a contradiction that  $\{f_{n_k}\}$  is a pointwise convergent subsequence that doesn't converge uniformly. Let  $f$  be the limit of  $f_{n_k}$ . There exists some  $\epsilon$  and some subsequence  $\{f_{n_{k_l}}\}$  such that for all  $l$  there exists some  $x_l \in X$  such that

$$d\left(f_{n_{k_l}}(x_l), f(x_l)\right) \geq \epsilon$$

Clearly,  $\{f_{n_{k_l}}\}$  contain no uniformly convergent subsequence, a contradiction. ■

## 1.3 Tests for Sequence and Series

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### Abstract

This section prove some basic result on sequence and series, which will be heavily used in [next section on analytic functions](#). Although written in an almost glossary form, we present the Theorems in a structural order based on the necessity of notion of absolute convergence and limit superior. Note that in this section,  $z, v, w$  always represent complex numbers, and  $a, b, c$  always represent real numbers.

---

**Theorem 1.3.1. (Weierstrass M-test)** Given sequences  $f_n : X \rightarrow \mathbb{C}$ , and suppose

$$\forall n \in \mathbb{N}, \forall x \in X, |f_n(x)| \leq M_n$$

Then

$$\sum_{n=1}^{\infty} M_n \text{ converge} \implies \sum_{n=1}^{\infty} f_n \text{ uniformly converge}$$

*Proof.* The proof follows from noting

$$\forall x \in X, \left| \sum_{k=m}^n f_k(x) \right| \leq \sum_{k=m}^n |f_k(x)| \leq \sum_{k=m}^n M_k$$

■

Note that in our proof of [Weierstrass M-test](#), we reduce the proof for uniform convergence into uniform Cauchy, which is a technique we shall also use later in [Abel's test for uniform convergence](#). We now prove [summation by part](#), which is a result hold in all fields, and is the essence of the proof of [Dirichlet's test](#) and [Abel's test for uniform convergence](#).

**Theorem 1.3.2. (Summation by Part)**

$$\begin{aligned} f_n g_n - f_m g_m &= \sum_{k=m}^{n-1} f_k \Delta g_k + g_k \Delta f_k + \Delta f_k \Delta g_k \\ &= \sum_{k=m}^{n-1} f_k \Delta g_k + g_{k+1} \Delta f_k \end{aligned}$$

*Proof.* The proof follows induction which is based on

$$f_m g_m + f_m \Delta g_m + g_m \Delta f_m + \Delta f_m \Delta g_m = f_{m+1} g_{m+1}$$

■



**Theorem 1.3.3. (Dirichlet's Test)** Suppose

- (a)  $a_n \rightarrow 0$  monotonically.
- (b)  $\sum_{n=1}^N z_n$  is bounded.

We have

$$\sum a_n z_n \text{ converge}$$

*Proof.* Define  $Z_n \triangleq \sum_{n=1}^N z_n$  and let  $M$  bound  $|Z_n|$ . Using **summation by part** by letting  $f_k = a_k$  and  $g_k = Z_{k-1}$ , we have

$$\begin{aligned} \left| \sum_{k=m}^n a_k z_k \right| &= \left| a_{n+1} Z_n - a_m Z_{m-1} - \sum_{k=m}^n Z_k (a_{k+1} - a_k) \right| \\ &\leq |a_{n+1} Z_n| + |a_m Z_{m-1}| + \left| \sum_{k=m}^n Z_k (a_{k+1} - a_k) \right| \\ (\because a_n \text{ is monotone}) \quad &\leq M \left( |a_{n+1}| + |a_m| + |a_{n+1} - a_m| \right) \end{aligned}$$

■

**Theorem 1.3.4. (Dirichlet's test for uniform convergence)** Let  $X$  be a metric space and  $\{f_n\}, \{g_n\}$  be two sequence of function that maps  $X$  into  $\mathbb{C}$ . If

- (i)  $f_n \searrow 0$  uniformly.
- (ii)  $\sum g_n$  is uniformly bounded.

then

$$\sum f_n g_n \text{ uniformly converges}$$

**Theorem 1.3.5. (Abel's Test for Uniform Convergence)** Suppose  $g_n : X \rightarrow \mathbb{R}$  is a uniformly bounded pointwise monotone sequence. Then given a sequence  $f_n : X \rightarrow \mathbb{R}$ ,

$$\sum f_n \text{ uniformly converge} \implies \sum f_n g_n \text{ uniformly converge}$$

*Proof.* Define  $R_n \triangleq \sum_{k=n}^{\infty} f_k$ . Let  $M$  uniformly bound  $g_n$ . Because  $R_n \rightarrow 0$  uniformly, we can let  $N$  satisfy

$$\forall n \geq N, \forall x \in X, |R_n(x)| < \frac{\epsilon}{3M}$$

Then for all  $n, m \geq N$ , using **summation by part**, we have

$$\begin{aligned}
\left| \sum_{k=m}^n f_k g_k \right| &= \left| \sum_{k=m}^n g_k \Delta R_k \right| \\
&\leq |R_{n+1} g_{n+1}| + |R_m g_m| + \sum_{k=m}^n |R_{k+1} \Delta g_k| \\
(\because g_n \text{ is pointwise monotone}) \quad &\leq |R_{n+1} g_{n+1}| + |R_m g_m| + \frac{\epsilon}{3M} |g_{n+1} - g_m| \leq \epsilon
\end{aligned}$$

■

Although the proofs of **Dirichlet's test** and **Abel's test for uniform convergence** are quite similar, one should note that the "ways" **summation by part** is applied are slightly different, as one use  $R_n \triangleq \sum_{k=n}^{\infty} f_k$  instead of  $\sum_{k=1}^n f_k$  like  $Z_n \triangleq \sum_{j=1}^n z_j$ . As corollaries of **Dirichlet's test**, one have the famous **alternating series test** and **Abel's test for complex series**.

**Theorem 1.3.6. (Abel's Test for Complex Series)** Suppose

- (a)  $\sum z_n$  converge.
- (b)  $b_n$  is a bounded monotone sequence.

We have

$$\sum z_n b_n \text{ converge}$$

*Proof.* Denote  $B \triangleq \lim_{n \rightarrow \infty} b_n$ . By **Dirichlet's Test**, we know  $\sum z_n (b_n - B)$  converge. The proof now follows form noting

$$\sum z_n b_n = \sum z_n (b_n - B) + B \sum z_n$$

■

We now introduce the idea of absolute convergence, which we shall use throughout the remaining of the section. By a **permutation**  $\sigma : E \rightarrow E$  on some set  $E$ , we merely mean  $\sigma$  is a bijective function. We say  $\sum z_n$  **absolutely converge** if  $\sum |z_n|$  converge, and say  $\sum z_n$  **unconditionally converge** if for all permutation  $\sigma : \mathbb{N} \rightarrow \mathbb{N}$ , the series  $\sum z_{\sigma(n)}$  converge and converge to the same value.

**Theorem 1.3.7. (Absolutely Convergent Series Unconditionally Converge)**

$$\sum z_n \text{ absolutely converge} \implies \sum_9 z_n \text{ unconditionally converge}$$

*Proof.* The fact  $\sum z_n$  converge follows from noting

$$\left| \sum_{k=n}^m z_k \right| \leq \sum_{k=n}^m |z_k| \leq \sum_{k=n}^{\infty} |z_k|$$

Now, fix  $\epsilon$  and permutation  $\sigma$ . Let  $N_1$  and  $N_2$  satisfy

$$\sum_{n=N_1}^{\infty} |z_n| < \frac{\epsilon}{2} \text{ and } \left| \sum_{n=N}^{\infty} z_n \right| < \frac{\epsilon}{2} \text{ for all } N > N_2$$

Let  $M \triangleq \max \{N_1, N_2\}$ . Observe that for all  $N > \max_{1 \leq r \leq M} \sigma^{-1}(r)$ , we have

$$\left| \sum z_n - \sum_{n=1}^N z_{\sigma(n)} \right| \leq \left| \sum_{n=M+1}^{\infty} z_n \right| + \sum_{n=M+1}^{\infty} |z_n| < \epsilon$$

■

**Theorem 1.3.8. (Riemann Rearrangement Theorem)** If  $\sum a_n$  converge but not absolutely, then for each  $L \in \overline{\mathbb{R}}$ , there exists a permutation  $\sigma$  such that

$$\sum a_{\sigma(n)} = L$$

*Proof.* Define  $a_n^+$  and  $a_n^-$  by

$$a_n^+ \triangleq \max \{a_n, 0\} \text{ and } a_n^- \triangleq \min \{a_n, 0\}$$

Because

$$\sum (a_n^+ + a_n^-) \text{ converge but } \sum (a_n^+ - a_n^-) = \infty$$

We know

$$\sum a_n^+ = \sum (-a_n^-) = \infty$$

WOLG,<sup>2</sup> fix  $L \in \mathbb{R}$  and suppose  $a_n \neq 0$  for all  $n$ . Let  $A = B = L$ , and let two increasing sequence  $\sigma^+, \sigma^- : \mathbb{N} \rightarrow \mathbb{N}$  satisfy

$$\sigma^+(k+1) = \min \{n \in \mathbb{N} : a_n > 0 \text{ and } n > \sigma^+(k)\}$$

---

<sup>2</sup>The proofs for  $L = \infty$  or  $-\infty$  are similar.

and similar for  $\sigma^-$ . Now, recursively define  $p_k, q_k$  by

$$p_1 \text{ is the smallest number such that } \sum_{n=1}^{p_1} a_{\sigma^+(n)} \geq A \quad (1.2)$$

$$q_1 \text{ is the smallest number such that } \sum_{n=1}^{p_1} a_{\sigma^+(n)} + \sum_{n=1}^{q_1} a_{\sigma^-(n)} \leq B \quad (1.3)$$

$$p_{k+1} \text{ is the smallest number such that } \sum_{n=1}^{p_{k+1}} a_{\sigma^+(n)} + \sum_{n=1}^{q_k} a_{\sigma^-(n)} \geq A \quad (1.4)$$

$$q_{k+1} \text{ is the smallest number such that } \sum_{n=1}^{p_{k+1}} a_{\sigma^+(n)} + \sum_{n=1}^{q_{k+1}} a_{\sigma^-(n)} \leq B \quad (1.5)$$

We then define  $\sigma$  by

$$\sigma^+(1), \dots, \sigma^+(p_1), \sigma^-(1), \dots, \sigma^-(q_1), \sigma^+(p_1 + 1), \dots, \sigma^+(p_2), \sigma^-(q_1 + 1), \dots, \sigma^-(q_2), \dots$$

It then follows from  $a_n \rightarrow 0$  that  $\sum a_{\sigma(n)} = L$ . ■

Note that the method we deploy in the proof of **Riemann rearrangement Theorem** can be used to control the sequence to have arbitrary large set of subsequential limits by modifying the number of  $A, B$  in **Equation (4.1), (4.2), (4.3) and (4.4)**.

Using **Riemann rearrangement Theorem** and equation

$$\max_{1 \leq r \leq d} |x_n| \leq |\mathbf{x}| \leq \sum_{r=1}^d |x_r|$$

we can now generalize and strengthen **Theorem 1.3.7** to

$$\begin{aligned} \sum \mathbf{x}_n \text{ absolutely converge} &\iff \sum_n x_{n,r} \text{ absolutely converge for all } r \\ &\iff \sum_n x_{n,r} \text{ unconditionally converge for all } r \\ &\iff \sum \mathbf{x}_n \text{ unconditionally converge} \end{aligned}$$

With this in mind, we can now well state the **Fubini's Theorem for Double Series**.

**Theorem 1.3.9. (Fubini's Theorem for Double Series)** If

$$\sum_n \sum_k |z_{n,k}| \text{ converge}$$

Then

$$\sum_{n,k} |z_{n,k}| \text{ converge and } \sum_{n,k} z_{n,k} = \sum_n \sum_k z_{n,k} = \sum_k \sum_n z_{n,k}$$

*Proof.* The fact  $\sum z_{n,k}$  absolutely converge follow from

$$\sum_{n=1}^N \sum_{k=1}^N |z_{n,k}| \leq \sum_n \sum_k |z_{n,k}| \text{ for all } N$$

WOLG, it remains to prove

$$\sum_{n,k} z_{n,k} = \sum_n \sum_k z_{n,k}$$

Because  $\sum_n \sum_k |z_{n,k}|$  converge, we can reduce the problem into proving the same statement for nonnegative series<sup>3</sup>  $\sum a_{n,k}$ :

$$\sum_n \sum_k |a_{n,k}| \text{ converge } \implies \sum_{n,k} a_{n,k} = \sum_n \sum_k a_{n,k}$$

Because

$$\sum_{n=1}^N \sum_{k=1}^N a_{n,k} \leq \sum_{n=1}^N \sum_k a_{n,k} \leq \sum_n \sum_k a_{n,k} \text{ for all } N$$

we see

$$\sum_{n,k} a_{n,k} \leq \sum_n \sum_k a_{n,k}$$

It remains to prove

$$\sum_{n,k} a_{n,k} \geq \sum_n \sum_k a_{n,k}$$

Fix  $N$  and  $\epsilon$ . We reduce the problem into proving

$$\sum_{n,k} a_{n,k} \geq \sum_{n=1}^N \sum_k a_{n,k} - \epsilon$$

---

<sup>3</sup>By splitting into real and imaginary parts, we can reduce the problem into proving the same statement for real series. Then, we can reduce the problem into proving the same statement for nonnegative series by proving for both the series  $\sum \max \{a_{n,k}, 0\}$  and  $\sum \min \{a_{n,k}, 0\}$

Let  $K$  satisfy

$$\text{For all } 1 \leq n \leq N, \quad \sum_{k=K+1}^{\infty} a_{n,k} < \frac{\epsilon}{N}$$

It then follows

$$\sum_{n,k} a_{n,k} \geq \sum_{n=1}^N \sum_{k=1}^K a_{n,k} \geq \sum_{n=1}^N \sum_k a_{n,k} - \epsilon \quad (\text{done})$$

■

**Example 1 (Counter-Example for Fubini's Theorem for Double Series)**

$$a_{n,k} \triangleq \begin{cases} 1 & \text{if } n = k \\ -1 & \text{if } n = k + 1 \\ 0 & \text{if otherwise} \end{cases}$$

$$\sum |a_{n,k}| = \infty \text{ and } \sum_n \sum_k a_{n,k} = 1 \text{ and } \sum_k \sum_n a_{n,k} = 0$$

**Theorem 1.3.10. (Merten's Theorem for Cauchy Product)** Suppose

(a)  $\sum_{n=0}^{\infty} z_n$  converge absolutely to  $Z$ .

(b)  $\sum_{n=0}^{\infty} v_n = V$

(c)  $w_n = \sum_{k=0}^n z_k v_{n-k}$

Then we have

$$\sum_{n=0}^{\infty} w_n = ZV$$

*Proof.* We prove

$$\left| V \sum_{n=0}^N z_n - \sum_{n=0}^N w_n \right| \rightarrow 0 \text{ as } N \rightarrow \infty$$

Compute

$$\begin{aligned} V \sum_{n=0}^N z_n - \sum_{n=0}^N w_n &= \sum_{n=0}^N z_n \left( V - \sum_{k=0}^{N-n} v_k \right) \\ &= \sum_{n=0}^N z_n \sum_{k=N-n+1}^{\infty} v_k \end{aligned}$$

Because  $\sum_{k=n}^{\infty} v_k \rightarrow 0$  as  $n \rightarrow \infty$ , we know there exists  $M$  such that

$$\left| \sum_{k=n}^{\infty} v_k \right| < M \text{ for all } n$$

Let  $N_0$  satisfy

$$\sum_{n=N_0+1}^{\infty} |z_n| < \frac{\epsilon}{2M}$$

Let  $N_1 > N_0$  satisfy

$$\left| \sum_{k=N-N_0+1}^{\infty} v_k \right| < \frac{\epsilon}{2(N_0+1) \sum_n |z_n|} \text{ for all } N > N_1$$

Now observe that for all  $N > N_1$

$$\left| \sum_{n=0}^N z_n \left( \sum_{k=N-n+1}^{\infty} v_k \right) \right| \leq \sum_{n=0}^{N_0} |z_n| \left| \sum_{k=N-n+1}^{\infty} v_k \right| + \sum_{n=N_0+1}^N |z_n| \left| \sum_{k=N-n+1}^{\infty} v_k \right| < \epsilon \text{ (done)}$$

■

We first define the **limit superior** by

$$\limsup_{n \rightarrow \infty} a_n \triangleq \lim_{n \rightarrow \infty} \left( \sup_{k \geq n} a_k \right)$$

Note that  $\limsup_{n \rightarrow \infty} a_n$  must exist because  $(\sup_{k \geq n} a_k)_n$  is a decreasing sequence.

**Theorem 1.3.11. (Equivalent Definition for Limit Superior)** If we let  $E$  be the set of subsequential limits of  $a_n$

$$E \triangleq \left\{ L \in \overline{\mathbb{R}} : L = \lim_{k \rightarrow \infty} a_{n_k} \text{ for some } n_k \right\}$$

The set  $E$  is non-empty and

$$\max E = \limsup_{n \rightarrow \infty} a_n$$

*Proof.* Let  $n_1 \triangleq 1$ . Recursively, because

$$\sup_{j \geq n_k} a_j \geq \limsup_{n \rightarrow \infty} a_n > \limsup_{n \rightarrow \infty} a_n - \frac{1}{k} \text{ for each } k$$

We can let  $n_{k+1}$  be the smallest number such that

$$a_{n_{k+1}} > \limsup_{n \rightarrow \infty} a_n - \frac{1}{k}$$

It is straightforward to check  $a_{n_k} \rightarrow \limsup_{n \rightarrow \infty} a_n$  as  $k \rightarrow \infty$ . Note that no subsequence can converge to  $\limsup_{n \rightarrow \infty} a_n + \epsilon$  because there exists  $N$  such that  $\sup_{k \geq N} a_k < \limsup_{n \rightarrow \infty} a_n + \epsilon$ . ■

We can now state the **limit comparison test** as follows. Given a positive sequence  $b_n$ ,

$$\begin{aligned} \limsup_{n \rightarrow \infty} \frac{|z_n|}{b_n} \in \mathbb{R} \text{ and } \sum b_n \text{ converge} &\implies \sum z_n \text{ absolutely converge} \\ \liminf_{n \rightarrow \infty} \frac{b_n}{|a_n|} > 0 \text{ and } \sum |a_n| \text{ diverge} &\implies \sum b_n \text{ diverge} \end{aligned}$$

### Theorem 1.3.12. (Geometric Series)

$$|z| < 1 \implies \sum_{n=0}^{\infty} z^n = \frac{1}{1-z}$$

*Proof.* The proof follows from noting

$$(1-z) \sum_{n=0}^N z^n = 1 - z^{N+1} \rightarrow 1 \text{ as } N \rightarrow \infty$$

### Theorem 1.3.13. (Ratio and Root Test)

$$\begin{aligned} \limsup_{n \rightarrow \infty} \sqrt[n]{|z_n|} < 1 \text{ or } \limsup_{n \rightarrow \infty} \left| \frac{z_{n+1}}{z_n} \right| < 1 &\implies \sum z_n \text{ absolutely converge} \\ \liminf_{n \rightarrow \infty} \sqrt[n]{|z_n|} > 1 \text{ or } \liminf_{n \rightarrow \infty} \left| \frac{z_{n+1}}{z_n} \right| > 1 &\implies \sum z_n \text{ diverge} \end{aligned}$$

*Proof.* The convergent part follows from comparison to an appropriate geometric series and the diverge part follows from noting  $|z_n|$  does not converge to 0. ■

### Theorem 1.3.14. (Root Test is Stronger Than Ratio Test)

$$\liminf_{n \rightarrow \infty} \left| \frac{z_{n+1}}{z_n} \right| \leq \liminf_{n \rightarrow \infty} \sqrt[n]{|z_n|} \leq \limsup_{n \rightarrow \infty} \sqrt[n]{|z_n|} \leq \limsup_{n \rightarrow \infty} \left| \frac{z_{n+1}}{z_n} \right|$$



*Proof.* Fix  $\epsilon$  and WLOG suppose  $\liminf_{n \rightarrow \infty} \left| \frac{z_{n+1}}{z_n} \right| > 0$ . We prove

$$\liminf_{n \rightarrow \infty} \sqrt[n]{|z_n|} \geq \liminf_{n \rightarrow \infty} \left| \frac{z_{n+1}}{z_n} \right| - \epsilon$$

Let  $\alpha \in \mathbb{R}$  satisfy

$$\liminf_{n \rightarrow \infty} \left| \frac{z_{n+1}}{z_n} \right| - \epsilon < \alpha < \liminf_{n \rightarrow \infty} \left| \frac{z_{n+1}}{z_n} \right|$$

Let  $N$  satisfy

$$\text{For all } n \geq N, \left| \frac{z_{n+1}}{z_n} \right| > \alpha$$

We then see

$$\sqrt[N+n]{|z_{N+n}|} \geq \sqrt[N+n]{|z_N|} \alpha^n \rightarrow \alpha \text{ as } n \rightarrow \infty \text{ (done)}$$

The proof for the other side is similar. ■

**Theorem 1.3.15. (Root Test Trick)** For all  $k \in \mathbb{N}$

$$\limsup_{n \rightarrow \infty} |z_{n+k}|^{\frac{1}{n}} = \limsup_{n \rightarrow \infty} |z_n|^{\frac{1}{n}}$$

*Proof.* Skipped. ■

Lastly, we prove **Cauchy's condensation Test**, whose existence is almost solely for investigating **p-Series**.

**Theorem 1.3.16. (Cauchy's Condensation Test)** Suppose  $a_n \searrow 0$ . We have

$$\sum_{n=0}^{\infty} 2^n a_{2^n} \text{ converge} \iff \sum_{n=1}^{\infty} a_n \text{ converge}$$

*Proof.* Observe that for all  $N \in \mathbb{N}$

$$\sum_{n=0}^N 2^n a_{2^n} \geq \sum_{n=1}^{2^{N+1}-1} a_n$$

and

$$\begin{aligned} a_1 + \frac{1}{2} \sum_{n=1}^K 2^n a_{2^n} &= a_1 + a_2 + 2a_4 + 4a_8 + \cdots + 2^{K-1} a_{2^K} \\ &\leq a_1 + a_2 + a_3 + a_4 + a_5 + a_6 + a_7 + a_8 + \cdots + a_{2^{K-1}+1} + \cdots + a_{2^K-1} + a_{2^K} \\ &\leq \sum_{n=1}^{\infty} a_n \end{aligned}$$

**Theorem 1.3.17. (p-Series)**

$$\sum_{n=1}^{\infty} \frac{1}{n^p} \text{ converge} \iff p > 1$$

*Proof.* Observe that

$$\sum_{n=0}^{\infty} 2^n \frac{1}{(2^n)^p} = \sum_{n=0}^{\infty} (2^{1-p})^n$$

The result then follows from **Cauchy's Condensation Test** and **geometric series**. ■

## 1.4 Analytic functions

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### Abstract

This section introduces the concept of analytic functions and proves some of their basic properties, including the **Identity Theorem**. We will rely on the tools developed in **the previous section on sequences and series**. Note that throughout this section,  $z$  will always denote a complex number.

---

In this section, by a **power series**, we mean a pair  $(z_0, c_n)$  where  $z_0 \in \mathbb{C}$  is called the **center** of power series, and  $c_n \in \mathbb{C}$  are the coefficients sequence. By **radius of convergence**, we mean a unique  $R \in \mathbb{R}_0^+ \cup \infty$  such that

$$\sum_{n=0}^{\infty} c_n (z - z_0)^n \begin{cases} \text{converge absolutely} & \text{if } |z - z_0| < R \\ \text{diverge} & \text{if } |z - z_0| > R \end{cases}$$

Such  $R$  always exist (and is unique, the uniqueness can be checked without computing the actual value of  $R$ ) and is exactly

$$R = \frac{1}{\limsup_{n \rightarrow \infty} \sqrt[n]{c_n}} \quad (1.6)$$

This result is called **Cauchy-Hadamard Theorem** and is proved by applying **Root Test** to  $\sum c_n (z - z_0)^n$ . Note that Cauchy-Hadamard Theorem does not tell us whether a power series converges at points of boundary of disk of convergence. It require extra works to determine if the power series converge at boundary.

**Theorem 1.4.1. (Abel's Test for Power Series)** Suppose  $\sum a_n z^n$  has radius of convergence  $R$  and  $a_n R^n \rightarrow 0$  monotonically. Then

The power series  $\sum a_n z^n$  at least converge on  $\overline{D_R(0)} \setminus \{R\}$

*Proof.* Fix  $z \in \overline{D_R(0)} \setminus \{R\}$ . Note that

$$\left| \sum_{n=0}^N \frac{z^n}{R^n} \right| = \left| \frac{1 - (\frac{z}{R})^{N+1}}{1 - \frac{z}{R}} \right| \leq \frac{2}{|1 - \frac{z}{R}|} \text{ for all } N$$

It then follows from **Dirichlet's Test** that  $\sum a_n R^n (\frac{z}{R})^n$  converge. ■

Let  $q \in \mathbb{R}$  and

### Example 2 (Discussion of Convergence on Boundary)

$$f_q(z) \triangleq \sum_{n=0}^{\infty} n^q z^n$$

It is clear that  $f_q$  has convergence radius 1 for all  $q \in \mathbb{R}$ . For boundary, we have

$$\begin{cases} q < -1 \implies f_q \text{ converge on } S^1 \\ -1 \leq q < 0 \implies f_q \text{ converge on } S^1 \setminus \{1\} \\ 0 \leq q \implies f_q \text{ diverge on } S^1 \end{cases}$$

Note that

- (a) At  $z = 1$ , the discussion is just **p-series**.
- (b)  $n^q \searrow 0$  if and only if  $q < 0$ ; and if  $n^q \searrow 0$ , then the series converge on  $S^1 \setminus \{1\}$  by **Abel's test for power series**.

Notice that the fact  $\sum c_n(z - z_0)^n$  absolutely converge in  $D_R(z_0)$  implies the convergence is uniform on all  $\overline{D_{R-\epsilon}(z_0)}$  by **M-Test**. However, on  $D_R(z_0)$ , the convergence is not always uniform.

### Example 3 (Failure of Uniform Convergence on $D_R(z_0)$ )

$$f(z) = \sum_{n=0}^{\infty} z^n$$

Note  $R = 1$ . Use **Geometric series formula** to show  $f(z) = \frac{1}{1-z}$  on  $D_1(0)$ . It is then clear that  $f$  is unbounded on  $D_1(0)$  while all partial sums  $\sum_{k=0}^n z^k$  is bounded on  $D_1(0)$ .

We now introduce some terminologies. We say a complex function  $f$  is **analytic at**  $z_0 \in \mathbb{C}$  if  $f$  there exists a power series  $(z_0, c_n)$  whose convergence radius is greater than 0 and  $f$  agrees with  $\sum_{n=0}^{\infty} c_n(z - z_0)^n$  on  $D_R(z_0)$  for some  $R$ . It shall be quite clear that if  $f, g$  are both analytic at  $z \in \mathbb{C}$  with radius  $R_f \leq R_g$ , then by **Merten's Theorem for Cauchy product**,  $f + g$  and  $fg$  are analytic at  $z$  with radius at least  $R_f$ . We now investigate deeper into analytic functions.

**Theorem 1.4.2. (Term by Term Differentiation)** Given a power series  $(z_0, c_n)$  of

convergence radius  $R > 0$ , if we define  $f : D_R(z_0) \rightarrow \mathbb{C}$  by

$$f(z) \triangleq \sum_{n=0}^{\infty} c_n (z - z_0)^n$$

Then  $f$  is holomorphic on  $D_R(z_0)$  and its derivative at  $z_0$  is also a power series with radius of convergence  $R$

$$f'(z) = \sum_{n=0}^{\infty} (n+1) c_{n+1} (z - z_0)^n$$

*Proof.* Because  $(n+1)^{\frac{1}{n}} \rightarrow 1$ , we can use [Theorem 1.3.15](#) to deduce

$$\limsup_{n \rightarrow \infty} ((n+1) |c_{n+1}|)^{\frac{1}{n}} = \limsup_{n \rightarrow \infty} |c_{n+1}|^{\frac{1}{n}} = \limsup_{n \rightarrow \infty} |c_n|^{\frac{1}{n}}$$

which implies that the power series  $\sum_{n=0}^{\infty} (n+1) c_{n+1} (z - z_0)^n$  is of radius of convergence  $R$ . We now prove

$$f'(z) = \sum_{n=0}^{\infty} (n+1) c_{n+1} (z - z_0)^n \text{ on } D_R(z_0)$$

Define  $f_m : D_R(z_0) \rightarrow \mathbb{C}$  by

$$f_m(z) \triangleq \sum_{n=0}^m c_n (z - z_0)^n$$

Observe

- (a)  $f_m \rightarrow f$  pointwise on  $D_R(a)$
- (b)  $f'_m(z) = \sum_{n=0}^{m-1} (n+1) c_{n+1} (z - z_0)^n$  for all  $m$

Fix  $z \in D_R(z_0)$ . Proposition (b) allow us to reduce the problem into proving

$$f'(z) = \lim_{m \rightarrow \infty} f'_m(z) \text{ on } D_R(a) \tag{1.7}$$

Let  $z \in D_r(z_0)$  where  $r < R$ . With proposition (a) in mind, to show [Equation 1.7](#), we only have to prove  $f'_m$  uniformly converge on  $D_r(z_0)$ , which follows from [M-Test](#) and the fact that  $\sum_{n=0}^{\infty} (n+1) c_{n+1} (z - z_0)^n$  absolutely converge on  $D_R(z_0)$ . [\(done\)](#) ■

Suppose

$$f(z) \triangleq \sum_{n=0}^{\infty} c_n (z - z_0)^n$$

Now by repeatedly applying **Theorem 1.4.2**, we see

$$f^{(k)}(z) = \sum_{n=0}^{\infty} (n+k) \cdots (n+1) c_{n+k} (z - z_0)^n \text{ for all } k \in \mathbb{Z}_0^+ \quad (1.8)$$

This then give us

$$c_k = \frac{f^{(k)}(z_0)}{k!} \text{ for all } k \in \mathbb{Z}_0^+ \quad (1.9)$$

and

$$f(z) = f(z_0) + f'(z_0)(z - z_0) + \frac{f''(z_0)}{2!}(z - z_0)^2 + \cdots \text{ on } D_R(z_0) \quad (1.10)$$

**Equation 1.10** is often called the **Taylor expansion of  $f$  at  $z_0$** . Notably, **Equation 1.9** tell us that if  $f$  is constant 0, then  $c_n = 0$  for all  $n$ .

#### Example 4 (Smooth but not Analytic Function)

$$f(x) = \begin{cases} e^{\frac{-1}{x^2}} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

Use induction to show that

$$f^{(k)}(x) = P_k\left(\frac{1}{x}\right)e^{-(\frac{1}{x})^2} \quad \exists P_k \in \mathbb{R}[x^{-1}], \forall k \in \mathbb{Z}_0^+, \forall x \in \mathbb{R}^*$$

and again use induction to show that

$$f^{(k)}(0) = 0 \quad \forall k \in \mathbb{Z}_0^+$$

Now, with **Theorem 1.4.2**, we see that  $f$  is not analytic at 0.

#### Example 5 (Bump Function)

$$f(x) = \begin{cases} e^{\frac{-1}{1-x^2}} & \text{if } |x| < 1 \\ 0 & \text{otherwise} \end{cases}$$

Use the same trick (but more advanced) to show  $f$  is smooth, and note that  $f$  is not analytic at  $\pm 1$ .

Now, it comes an interesting question. Given a complex-valued function  $f$  analytic at  $z_0$  with radius  $R$ , and suppose  $z_1 \in D_R(z_0)$ .

- (a) Is  $f$  also analytic at  $z_1$ ?
- (b) What do we know about the radius of convergence of  $f$  at  $z_1$ ?
- (c) Suppose  $f$  is indeed analytic at  $z_1$ . It is trivial to see that the power series  $(z_0, c_{0;n})$  and  $(z_1, c_{1;n})$  must agree in the intersection of their convergence disks, and because  $f$  is given, we—by [Equation 1.9](#)—have already known the value of  $c_{1;n}$ . Can we verify that the power series  $(z_0, c_{0;n})$  and  $(z_1, c_{1;n})$  do indeed agree with each other on the common convergence interval?

[Taylor's Theorem for power series](#) give satisfying answers to these problems.

**Theorem 1.4.3. (Taylor's Theorem for Power Series)** Given a function  $f$  analytic at  $z_0$  with radius  $R$ , and suppose  $z_1 \in D_R(z_0)$ . Then

$$f(z) = \sum_{k=0}^{\infty} \frac{f^{(k)}(z_1)}{k!} (z - z_1)^k \text{ on } D_{R-|z_1-z_0|}(z_1)$$

*Proof.* WOLG, let  $z_0 = 0$ . Suppose  $z$  satisfy  $|z - z_1| < R - |z_1|$ . By [Equation 1.9](#), we can compute

$$\begin{aligned} f(z) &= \sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!} z^k \\ &= \sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!} (z - z_1 + z_1)^k \\ &= \sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!} \sum_{n=0}^k \binom{k}{n} (z - z_1)^n z_1^{k-n} \\ &= \sum_{k=0}^{\infty} \sum_{n=0}^k \frac{f^{(k)}(0)}{k!} \binom{k}{n} (z - z_1)^n z_1^{k-n} \end{aligned}$$

Note that

$$\begin{aligned}
\sum_{k=0}^{\infty} \left| \sum_{n=0}^{\infty} \frac{f^{(k)}(0)}{k!} \binom{k}{n} (z - z_1)^n z_1^{k-n} \right| &\leq \sum_{k=0}^{\infty} \sum_{n=0}^{\infty} \left| \frac{f^{(k)}(0)}{k!} \right| \binom{k}{n} |z - z_1|^n \cdot |z_1|^{k-n} \\
&= \sum_{k=0}^{\infty} \left| \frac{f^{(k)}(0)}{k!} \right| \sum_{n=0}^{\infty} \binom{k}{n} |z - z_1|^n \cdot |z_1|^{k-n} \\
&= \sum_{k=0}^{\infty} \left| \frac{f^{(k)}(0)}{k!} \right| (|z - z_1| + |z_1|)^k
\end{aligned}$$

is a convergent series, by **Cauchy-Hadamard Theorem** and  $|z - z_1| + |z_1| < R$ . Therefore, we can use **Fubini's Theorem for double series** to deduce

$$\begin{aligned}
\sum_{k=0}^{\infty} \sum_{n=0}^k \frac{f^{(k)}(0)}{k!} \binom{k}{n} (z - z_1)^n z_1^{k-n} &= \sum_{k=0}^{\infty} \sum_{n=0}^{\infty} \frac{f^{(k)}(0)}{k!} \binom{k}{n} (z - z_1)^n z_1^{k-n} \\
&= \sum_{n=0}^{\infty} \sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!} \binom{k}{n} (z - z_1)^n z_1^{k-n} \\
&= \sum_{n=0}^{\infty} \sum_{k=n}^{\infty} \frac{f^{(k)}(0)}{k!} \binom{k}{n} (z - z_1)^n z_1^{k-n} \\
&= \sum_{n=0}^{\infty} \left[ \sum_{k=n}^{\infty} \frac{f^{(k)}(0)}{k!} \binom{k}{n} z_1^{k-n} \right] (z - z_1)^n
\end{aligned}$$

We have reduced the problem into proving

$$\sum_{k=n}^{\infty} \frac{f^{(k)}(0)}{k!} \binom{k}{n} z_1^{k-n} = \frac{f^{(n)}(z_1)}{n!}$$

Because  $z_1$  is in  $D_R(0)$ , by **Equation 1.8** and **Equation 1.9**, we can compute

$$\begin{aligned}
f^{(n)}(z_1) &= \sum_{k=0}^{\infty} (k+n) \cdots (k+1) \cdot \frac{f^{(n+k)}(0)}{(n+k)!} z_1^k \\
&= \sum_{k=n}^{\infty} (k) \cdots (k-n+1) \cdot \frac{f^{(k)}(0)}{k!} \cdot z_1^{k-n} \\
&= \sum_{k=n}^{\infty} \frac{f^{(k)}(0)}{(k-n)!} z_1^{k-n}
\end{aligned}$$



We now have

$$\frac{f^{(n)}(z_1)}{n!} = \sum_{k=n}^{\infty} \frac{f^{(k)}(0)}{n!(k-n)!} z_1^{k-n} = \sum_{k=n}^{\infty} \frac{f^{(k)}(0)}{k!} \binom{k}{n} z_1^{k-n} \text{ (done)}$$

■

Lastly, to close this section, we prove the **Identity Theorem**, which is extremely useful in complex analysis.

**Theorem 1.4.4. (Identity Theorem)** Given two analytic complex-valued function  $f, g : D \rightarrow \mathbb{C}$  defined on some open connected  $D \subseteq \mathbb{C}$ , if  $f, g$  agree on some subset  $S \subseteq D$  such that  $S$  has a limit point in  $D$ , then  $f, g$  agree on the whole region  $D$ .

*Proof.* Define

$$T \triangleq \left\{ z \in D : f^{(k)}(z) = g^{(k)}(z) \text{ for all } k \geq 0 \right\}$$

Since  $D$  is connected, we can reduce the problem into proving  $T$  is non-empty, open and closed in  $D$ . Let  $c$  be a limit point of  $S$  in  $D$ . We first show

$$c \in T$$

Assume for a contradiction that  $c \notin T$ . Let  $m$  be the smallest integer such that  $f^{(m)}(c) \neq g^{(m)}(c)$ . We can write the Taylor expansion of  $f - g$  at  $c$  by

$$\begin{aligned} (f - g)(z) &= (z - c)^m \left[ \frac{(f - g)^{(m)}(c)}{m!} + \frac{(f - g)^{(m+1)}(c)}{(m+1)!} (z - c) + \dots \right] \\ &\triangleq (z - c)^m h(z) \end{aligned}$$

Clearly,  $h(c) \neq 0$ . Now, because  $h$  is continuous at  $c$  ( $h$  is a well-defined power series at  $c$  with radius greater than 0), we see  $h$  is non-zero on some  $B_\epsilon(c)$ , which is impossible, since  $(f - g) \equiv 0$  on  $S \setminus \{c\}$  implies  $h = 0$  on  $S \setminus \{c\}$ . (done)

Fix  $z \in T$ . Because  $f, g$  are analytic at  $z$  and  $f^{(k)}(z) = g^{(k)}(z)$  for all  $k$ , we see  $f - g$  is constant 0 on some open disk  $B_\epsilon(z)$ . We have proved that  $T$  is open. To see  $T$  is closed in  $D$ , one simply observe that

$$T = \bigcap_{k \geq 0} \left\{ z \in D : (f - g)^{(k)}(z) = 0 \right\}$$

and  $(f - g)^{(k)}$  is continuous on  $D$ . (done)

■

## 1.5 Inverse function theorem

Let  $X$  be a metric space. A **contraction**  $f : X \rightarrow X$  is a  $K$ -Lipschitz function such that  $K < 1$ .

**Example 1.5.1.** Let  $X \triangleq \mathbb{R}$ .  $f(x) \triangleq \frac{x}{2}$  is a  $\frac{1}{2}$ -contraction, and  $g(x) \triangleq x$  for no  $r \in [0, 1)$  is a  $r$ -contraction.

**Theorem 1.5.2. (Banach's fixed point theorem)** Let  $X$  be a metric space. Let  $f : X \rightarrow X$  be a contraction. Then  $f$  admits at most one fixed point. Let  $f^n : X \rightarrow X$  denote:

$$\overbrace{f \circ \cdots \circ f}^{n \text{ copies}} : X \rightarrow X$$

If  $X$  is moreover complete, then for all  $x \in X$ , the sequence  $\{f^n(x)\} \subseteq X$  converges, with limit being a (the unique) fixed point of  $f$ .

*Proof.* It is easy to see from the definition that a contraction can only have at most one fixed point from the definition. Let  $X$  be complete and  $x \in X$ . Because  $f$  as a Lipschitz function is continuous, we know that if  $\{f^n(x)\} \subseteq X$  converges, then its limit is a (the unique) fixed point of  $f$ :

$$f\left(\lim_{n \rightarrow \infty} f^n(x)\right) = \lim_{n \rightarrow \infty} f^{n+1}(x) = \lim_{n \rightarrow \infty} f^n(x)$$

It remains to prove  $\{f^n(x)\} \subseteq X$  converges. Because  $X$  is complete, we only have to prove  $\{f^n(x)\} \subseteq X$  is Cauchy. Indeed, given some  $n \in \mathbb{N}$ , we have:

$$d(f^n(x), f^{n+k}(x)) \leq K^k d(x, f^k(x)) \rightarrow 0$$

since  $d(x, f^k(x)) \rightarrow d(x, f(x))$ . ■

**Theorem 1.5.3. (Pointwise inverse function theorem)** Let  $U$  be some neighborhood of  $\mathbf{a} \in \mathbb{R}^n$ . If differentiable  $\mathbf{f} : U \rightarrow \mathbb{R}^n$  is  $C^1$  at  $\mathbf{a}$  and has full rank Jacobian at  $\mathbf{a}$ , then:

- (i)  $\mathbf{f}$  maps some  $B_\delta(\mathbf{a}) \subseteq U$  injectively to some open subset of  $\mathbb{R}^n$ .
- (ii) Moreover, the local inverse of  $\mathbf{f}$  is differentiable at  $\mathbf{f}(\mathbf{a})$ .

*Proof.* Skipped. ■

**Corollary 1.5.4. (Inverse function theorem)** Let  $U$  be some neighborhood of  $\mathbf{a}$ . If  $C^1$  function  $\mathbf{f} : U \rightarrow \mathbb{R}^n$  has full rank Jacobian at  $\mathbf{a}$ , then  $\mathbf{f}$  forms a  $C^1$ -diffeomorphism between some neighborhood  $W \ni \mathbf{a}$  contained by  $U$ , and  $\mathbf{f}(W)$ .

*Proof.* Because  $GL_n(\mathbb{R}) \subseteq M_n(\mathbb{R})$  is open and  $\mathbf{f}$  is  $C^1$ , we know there exists some neighborhood  $U' \ni \mathbf{a}$  contained by  $U$  such that  $\mathbf{f}$  has full rank Jacobian everywhere in  $U'$ . Applying **pointwise inverse function theorem** to every point in  $U'$ , we get an injective  $C^1$ -function  $\mathbf{f} : W \rightarrow \mathbb{R}^n$  with differentiable inverse. To see  $(\mathbf{f})^{-1} : \mathbf{f}(W) \rightarrow W$  is also  $C^1$ , just observe:

$$\mathbf{y} \mapsto d\mathbf{f}^{-1}|_{\mathbf{y}} = (\mathbf{x} \mapsto (d\mathbf{f}|_{\mathbf{x}})^{-1}) \circ (\mathbf{y} \mapsto \mathbf{f}^{-1}(\mathbf{y}))$$

and recall that  $GL_n(\mathbb{R}) \rightarrow GL_n(\mathbb{R}), A \mapsto A^{-1}$  is continuous. ■

## 1.6 Implicit function theorem

**Theorem 1.6.1. (Implicit function theorem)** Let  $W \subseteq \mathbb{R}^n \times \mathbb{R}^m$  be open and  $(\mathbf{a}, \mathbf{b}) \in W$ . Let  $\mathbf{f}: W \rightarrow \mathbb{R}^n$  be a  $C^1$  function that makes the matrix:

$$\begin{pmatrix} \left. \frac{\partial f_1}{\partial x_1} \right|_{(\mathbf{a}, \mathbf{b})} & \cdots & \left. \frac{\partial f_1}{\partial x_n} \right|_{(\mathbf{a}, \mathbf{b})} \\ \vdots & \ddots & \vdots \\ \left. \frac{\partial f_n}{\partial x_1} \right|_{(\mathbf{a}, \mathbf{b})} & \cdots & \left. \frac{\partial f_n}{\partial x_n} \right|_{(\mathbf{a}, \mathbf{b})} \end{pmatrix}$$

invertible. Then, there exists a pair of neighborhoods  $U \ni (\mathbf{a}, \mathbf{b})$  and  $V \ni \mathbf{b}$  such that: For all  $\mathbf{y} \in V$  there exists a unique  $\mathbf{x} \in \mathbb{R}^n$  that satisfies both  $(\mathbf{x}, \mathbf{y}) \in U$  and  $\mathbf{f}(\mathbf{x}, \mathbf{y}) = \mathbf{f}(\mathbf{a}, \mathbf{b})$ . Moreover, if we define  $\mathbf{g}: V \rightarrow \mathbb{R}^n$  by setting

$$\mathbf{f}(\mathbf{g}(\mathbf{y}), \mathbf{y}) = \mathbf{f}(\mathbf{a}, \mathbf{b}), \quad \text{for all } \mathbf{y} \in V$$

then  $\mathbf{g}$  will be  $C^1$  and clearly make  $\mathbf{g}(\mathbf{b}) = \mathbf{a}$ .

*Proof.* Define  $\mathbf{F}: W \rightarrow \mathbb{R}^n \times \mathbb{R}^m$  by

$$\mathbf{F}(\mathbf{x}, \mathbf{y}) \triangleq (\mathbf{f}(\mathbf{x}, \mathbf{y}), \mathbf{y})$$

By **differentiability theorem**,  $\mathbf{F}$  is  $C^1$ . Clearly, the Jacobian of  $\mathbf{F}$  at  $(\mathbf{a}, \mathbf{b})$  is full rank. We may now apply **inverse function theorem** to find some smaller neighborhood  $U \subseteq W$  of  $(\mathbf{a}, \mathbf{b})$  such that:

- (I)  $\mathbf{F}$  is injective on  $U$ .
- (II)  $\mathbf{F}(U) \subseteq \mathbb{R}^n \times \mathbb{R}^m$  is open.
- (III)  $\mathbf{F}$  has full rank Jacobian everywhere in  $U$ .
- (IV) The local inverse  $\mathbf{F}^{-1}: \mathbf{F}(U) \rightarrow U$  is  $C^1$ .

Because  $\mathbf{F}(U)$  is open, we know

$$V \triangleq \{\mathbf{y} \in \mathbb{R}^m : (\mathbf{f}(\mathbf{a}, \mathbf{b}), \mathbf{y}) \in \mathbf{F}(U)\} \text{ is open}$$

We claim such  $U \ni (\mathbf{a}, \mathbf{b})$  and  $V \ni \mathbf{b}$  suffices. Fix  $\mathbf{y} \in V$ . We are required to show the existence and uniqueness of  $\mathbf{x} \in \mathbb{R}^n$  that satisfies both  $(\mathbf{x}, \mathbf{y}) \in U$  and  $\mathbf{f}(\mathbf{x}, \mathbf{y}) = \mathbf{f}(\mathbf{a}, \mathbf{b})$ .

By definition of  $V$ , we know  $(\mathbf{f}(\mathbf{a}, \mathbf{b}), \mathbf{y}) \in \mathbf{F}(U)$ . Therefore, there exist  $\mathbf{z} = (\mathbf{z}_1, \mathbf{z}_2) \in U$  such that  $\mathbf{F}(\mathbf{z}) = (\mathbf{f}(\mathbf{a}, \mathbf{b}), \mathbf{y})$ . By definition of  $\mathbf{F}$ ,  $\mathbf{z}_1$  is the  $\mathbf{x}$  we are looking for. Moreover, because  $\mathbf{F}$  is injective on  $U$ , we know  $\mathbf{z}_1$  is the only option. We have shown the existence and uniqueness of  $\mathbf{x}$ . It remains to prove that  $\mathbf{g}$  is  $C^1$ , which follows from **differentiability theorem** and that  $(\mathbf{g}(\mathbf{y}), \mathbf{y}) = \mathbf{F}^{-1}(\mathbf{f}(\mathbf{a}, \mathbf{b}), \mathbf{y})$  for all  $\mathbf{y} \in V$ . ■

Note that  $\mathbf{g}$  clearly satisfies:

$$\begin{pmatrix} \frac{\partial g_1}{\partial y_1} \Big|_{\mathbf{y}} & \cdots & \frac{\partial g_1}{\partial y_m} \Big|_{\mathbf{y}} \\ \vdots & \ddots & \vdots \\ \frac{\partial g_n}{\partial y_1} \Big|_{\mathbf{y}} & \cdots & \frac{\partial g_n}{\partial y_m} \Big|_{\mathbf{y}} \end{pmatrix} = - \begin{pmatrix} \frac{\partial f_1}{\partial x_1} \Big|_{(\mathbf{g}(\mathbf{y}), \mathbf{y})} & \cdots & \frac{\partial f_1}{\partial x_n} \Big|_{(\mathbf{g}(\mathbf{y}), \mathbf{y})} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1} \Big|_{(\mathbf{g}(\mathbf{y}), \mathbf{y})} & \cdots & \frac{\partial f_n}{\partial x_n} \Big|_{(\mathbf{g}(\mathbf{y}), \mathbf{y})} \end{pmatrix}^{-1} \cdot \begin{pmatrix} \frac{\partial f_1}{\partial y_1} \Big|_{(\mathbf{g}(\mathbf{y}), \mathbf{y})} & \cdots & \frac{\partial f_1}{\partial y_m} \Big|_{(\mathbf{g}(\mathbf{y}), \mathbf{y})} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial y_1} \Big|_{(\mathbf{g}(\mathbf{y}), \mathbf{y})} & \cdots & \frac{\partial f_n}{\partial y_m} \Big|_{(\mathbf{g}(\mathbf{y}), \mathbf{y})} \end{pmatrix}$$

for all  $\mathbf{y} \in V$ , which in particular manifest the fact that  $\mathbf{g}$  is  $C^1$ .

## 1.7 L'Hospital Rule

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### Abstract

This section state and prove the **L'Hospital Rule**. Note that although **L'Hospital Rule** is not really directly used in most results we teach in class, it is used in the proof of **Taylor's Theorem**.

---

**Theorem 1.7.1. (L'Hospital Rule)** Let  $I \subseteq \mathbb{R}$  be an open interval containing  $c$  and let  $f, g : I \rightarrow \mathbb{R}$  be two function continuous on  $I$  and differentiable on  $I$  everywhere except possibly at  $c$ , where

$$g'(x) \neq 0 \text{ for all } x \in I \setminus \{c\}$$

If  $\frac{f}{g}$  is indeterminate form, i.e.

$$\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} g(x) = L \text{ where } L \in \{0, \infty, -\infty\}$$

and

$$\lim_{x \rightarrow c} \frac{f'(x)}{g'(x)} \in \mathbb{R}$$

Then we have

$$\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \lim_{x \rightarrow c} \frac{f'(x)}{g'(x)} \quad (1.11)$$

*Proof.* Suppose  $I = (a, b)$ . Note that since  $g'(x) \neq 0$  on  $(c, b)$ , by **MVT**, we know there exists at most one  $x \in (c, b)$  such that  $g(x) = 0$ . With similar argument for  $(a, c)$ , we see that

$$g(x) \neq 0 \text{ on } (c - \epsilon, c + \epsilon) \setminus \{c\} \text{ for some } \epsilon$$

Therefore, WOLG, we can suppose  $g(x) \neq 0$  on  $I \setminus \{c\}$ . Define  $m, M : I \setminus \{c\} \rightarrow \mathbb{R}$  by

$$m(x) \triangleq \inf \frac{f'(t)}{g'(t)} \text{ and } M(x) \triangleq \sup \frac{f'(t)}{g'(t)} \text{ where } t \text{ ranges over value strictly between } x \text{ and } c$$

Because the value  $\frac{f'(t)}{g'(t)}$  converge at  $c$ , we can deduce

$$\lim_{x \rightarrow c} m(x) = \lim_{x \rightarrow c} M(x) = \lim_{x \rightarrow c} \frac{f'(x)}{g'(x)} \quad (1.12)$$

We now prove

the case when  $\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} g(x) = 0$

Fix  $x \in I \setminus \{c\}$ , and WOLOG suppose  $x < c$ . By Cauchy's MVT, we know that

$$m(x) \leq \frac{f(x) - f(y)}{g(x) - g(y)} \leq M(x) \text{ for all } y \in (x, c)$$

Note that  $g(x) \neq g(y)$  by MVT, since  $g' \neq 0$  on  $I \setminus \{c\}$ . It now follows from

$$\lim_{y \rightarrow c^-} \frac{f(x) - f(y)}{g(x) - g(y)} = \frac{f(x)}{g(x)}$$

that

$$m(x) \leq \frac{f(x)}{g(x)} \leq M(x)$$

The proof of Equation 1.11 then follows from Equation 1.12. (done)

We now prove

the case when  $\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} g(x) = \infty$

Again, fix  $x \in I \setminus \{c\}$ , and WOLOG suppose  $x < c$ . By Cauchy's MVT, we know that

$$m(x) \leq \frac{f(y) - f(x)}{g(y) - g(x)} \leq M(x) \text{ for all } y \in (x, c)$$

Note that  $g(x) \neq g(y)$  by MVT, since  $g' \neq 0$  on  $I \setminus \{c\}$ . It now follows

$$\frac{f(y) - f(x)}{g(y) - g(x)} = \frac{\frac{f(y)}{g(y)} - \frac{f(x)}{g(y)}}{1 - \frac{g(x)}{g(y)}} \rightarrow \lim_{x \rightarrow c} \frac{f(x)}{g(x)} \text{ as } y \rightarrow c^-$$

The proof of Equation 1.11 then follows from Equation 1.12. (done) ■

## 1.8 Lebesgue calculus

**Theorem 1.8.1. (Dominated convergence theorem)** Let  $E \subseteq X$  be measurable and  $f_n$  be a sequence of complex-valued function measurable on  $E$  that converges almost everywhere to some  $f$  on  $E$ . If there exists some  $\varphi \in L(E)$  such that  $|f_n| \leq \varphi$  almost everywhere for all  $n \in \mathbb{N}$ , then

$$\lim_{n \rightarrow \infty} \int_E |f - f_n| d\mathbf{x} = 0$$

*Proof.* Skipped. ■

**Theorem 1.8.2. (Change of variables formula)** Let  $X \subseteq \mathbb{R}^n$  be open,  $T : X \rightarrow \mathbb{R}^n$  be  $C^1$  and injective. For all measurable function  $f : \mathbb{R}^n \rightarrow [0, \infty]$ , we have:

$$\int_{T(X)} f d\mathbf{m} = \int_X (f \circ T) |J_T| d\mathbf{m}$$

*Proof.* Rudin RAC Theorem 7.26 ■

**Theorem 1.8.3. (Fubini's and Tonelli's theorem for double integral)** Let  $f : \mathbb{R}^{n+m} \rightarrow \mathbb{R}$  be measurable. If  $f$  satisfies one of the followings conditions:

- (I)  $f$  is integrable.
- (II)  $f$  is nonnegative.

then we have the formula:

$$\int_{\mathbb{R}^{n+m}} f(\mathbf{x}) d\mathbf{x} = \int_{\mathbb{R}^n} g(\mathbf{y}) d\mathbf{y}, \quad \text{where } g(\mathbf{y}) \triangleq \int_{\mathbb{R}^m} f(\mathbf{y}, \mathbf{z}) d\mathbf{z}$$

even if the integral may be  $+\infty$ .

*Proof.* Skipped. ■

**Theorem 1.8.4. (Measure-theoretic Feymann's trick)** Let  $X$  be an open subset of  $\mathbb{R}$  and  $\Omega$  a measure space. If  $f : X \times \Omega \rightarrow \mathbb{R}$  satisfies:

- (I)  $\Omega \rightarrow \mathbb{R} : t \mapsto f(x, t)$  is an integrable function for all  $x \in X$ .
- (II) For almost all  $t \in \Omega$ , the partial derivative  $\partial f / \partial x$  exists for all  $x \in X$ .
- (III) There is an integrable function  $\theta : \Omega \rightarrow \mathbb{R}$  such that for almost every  $t \in \Omega$  we have:

$$\left| \frac{\partial f}{\partial x}(x, t) \right| \leq \theta(t), \quad \text{for all } x \in X$$



Then,

$$\frac{d}{dx} \int_{\Omega} f(x, \omega) d\omega = \int_{\Omega} \frac{\partial f}{\partial x} d\omega, \quad \text{for all } x \in X$$

*Proof.* Skipped. ■

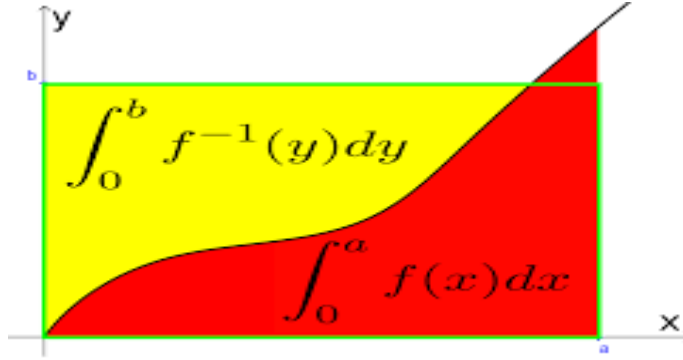
## 1.9 Young and Hölder inequalities

**Theorem 1.9.1. (A geometric estimation)** If  $\varphi : [0, \infty) \rightarrow \mathbb{R}$ , subject to  $\varphi(0) = 0$ , is continuous and strictly increasing, then for all  $a, b \in [0, \infty]$ , we have:

$$ab \leq \int_0^a \varphi(x) dx + \int_0^b \varphi^{-1}(y) dy$$

where  $\varphi^{-1} : [0, \infty) \rightarrow \mathbb{R}$  is the inverse function of  $\varphi$ . Moreover, equality holds if and only if  $b = \varphi(a)$ .

*Proof.* The proof is straightforward to come up after an observation of the graph:



■

We say a pair of numbers  $p, q \in [1, \infty]$  are **exponentially conjugated** if

$$\frac{1}{p} + \frac{1}{q} = 1$$

**Theorem 1.9.2. (Young's inequality for product)** If  $a, b \in [0, \infty)$  and  $p, q \in (1, \infty)$  is an exponential conjugated pair, then:

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}$$

Moreover, equality holds if and only if  $a^p = b^q$ .

*Proof.* The proof follows from setting  $\varphi(x) \triangleq x^{p-1}$  in the **above geometric estimation**. ■

Fix  $1 \leq p < \infty$ . Let  $E \subseteq \mathbb{R}$  be measurable, and define for all  $\mathbb{R}$ -valued function  $f$  measurable on  $E$  that

$$\|f\|_p \triangleq \left( \int_E |f|^p \right)^{\frac{1}{p}}$$

and that

$$\|f\|_\infty \triangleq \inf \{M \in \mathbb{R}_0^+ : |f| \leq M \text{ a.e. on } E\}$$

Clearly, for all  $1 \leq p \leq \infty$ , the space  $L^p(E)$  of  $\mathbb{R}$ -valued functions  $f$  measurable on  $E$  that satisfies  $\|f\|_p < \infty$  forms a  $\mathbb{R}$ -vector space. Clearly, for each  $p$ , the  **$p$ -norm**  $\|\cdot\|_p$  form a semi-norm on  $L^p(E)$ .

**Theorem 1.9.3. ( $L^{p+\epsilon}$  implies  $L^p$  on finite domain)** Let  $1 \leq p < \infty$  and  $E \subseteq \mathbb{R}$ . Then:

$$|E| < \infty \text{ and } 0 \leq N \leq \infty \implies L^{p+N}(E) \subseteq L^p(E)$$

*Proof.* The  $N = \infty$  case is clear. Let  $N < \infty$  and let  $f \in L^{p+N}(E)$ . The proof then follows from direct estimation based on splitting:

$$E = \{x \in E : |f(x)| \leq 1\} \cup \{x \in E : |f(x)| \geq 1\}$$

■

**Theorem 1.9.4. (Hölder's inequality)** Let  $p, q \in [1, \infty]$  be an exponential conjugated pair,  $E \subseteq \mathbb{R}$  be measurable, and  $f, g$  be two  $\mathbb{R}$ -valued functions measurable on  $E$ . Then:

$$\|fg\|_1 \leq \|f\|_p \|g\|_q$$

If moreover  $p > 1$ , then the equality hold if and only if there exists  $c \in \mathbb{R}$  such that  $|f|^p = c|g|^q$  a.e. on  $E$ .

*Proof.* The proof relies on *normalization*. Before such, we first have to handle two special cases: the case of  $p = 1$  and the case of  $\|f\|_p = 0$ . Both of them are easy. The latter follows from an observation that  $\|f\|_p = 0 \implies fg = 0$  almost everywhere, which follows from definition of Lebesgue integral.

We may from now on suppose  $1 < p < \infty$  and  $\|f\|_p \neq 0 \neq \|g\|_q$ . Define  $\hat{f}, \hat{g}$  by

$$\hat{f} \triangleq \frac{f}{\|f\|_p} \quad \text{and} \quad \hat{g} \triangleq \frac{g}{\|g\|_q}$$

The proof then follows from **Young's inequality**:

$$\|\hat{f} \cdot \hat{g}\|_1 \leq \int_E \frac{|\hat{f}|^p}{p} + \frac{|\hat{g}|^q}{q} = \frac{1}{p} + \frac{1}{q} = 1$$

■

Let  $E \subseteq \mathbb{R}$  be measurable, one may check that  $\|\cdot\|_2$  satisfies **parallelogram law** and is induced by the obvious inner product:

$$\langle f, g \rangle \triangleq \int_E f \bar{g}$$

Therefore, **Hölder's inequality** become **Cauchy-Schwarz inequality** of  $L^2$ -space.

## 1.10 Lebesgue's criteria for proper Riemann integrability

Let  $f : [a, b] \rightarrow \mathbb{R}$  be bounded. We say  $f$  is **proper Riemann integrable** on  $[a, b]$  if for all  $\epsilon$ , there exists a **partition**  $P = \{a = x_0 < \cdots < x_n = b\}$  such that

$$U(P) - L(P) \leq \epsilon$$

where

$$U(P) \triangleq \sum_{k=1}^n \left( \sup_{x \in I_k} f(x) \right) (x_k - x_{k-1}), \quad \text{where } I_k = [x_{k-1}, x_k]$$

$$\text{and } L(P) \triangleq \sum_{k=1}^n \left( \inf_{x \in I_k} f(x) \right) (x_k - x_{k-1}), \quad \text{where } I_k = [x_{k-1}, x_k]$$

Given some subset  $I \subseteq [a, b]$ , the **oscillation** of  $f$  on  $I$  is:

$$\text{osc}(f, I) \triangleq \sup_{x, y \in I} |f(x) - f(y)|$$

Because we clearly have:

$$\sup_{x \in I} f(x) - \inf_{x \in I} f(x) = \text{osc}(f, I)$$

we actually have:

$$U(P) - L(P) = \sum_{k=1}^n \text{osc}(f, I_k)(x_k - x_{k-1}), \quad \text{where } I_k = [x_{k-1}, x_k] \quad (1.13)$$

For all  $x \in [a, b]$ , we define the **oscillation** of  $f$  at  $x$  to be

$$\text{osc}(f, x) \triangleq \inf_{\delta > 0} \text{osc}(f, [x - \delta, x + \delta] \cap [a, b])$$

Clearly,  $f : [a, b] \rightarrow \mathbb{R}$  is continuous at  $x$  if and only if  $\text{osc}(f, x) = 0$ . This observation together with **characterization 1.13** is the moral reason we see a relationship between continuity and proper Riemann integrability. We write:

$$D(\epsilon) \triangleq \{x \in [a, b] : \text{osc}(f, x) \geq \epsilon\}$$

**Theorem 1.10.1. (Lebesgue's criteria for proper Riemann integrability)** Let  $f : [a, b] \rightarrow \mathbb{R}$  be bounded, and define

$$D_f \triangleq \{x \in [a, b] | f : [a, b] \rightarrow \mathbb{R} \text{ is discontinuous at } x\}$$

We have

$$f \text{ is proper Riemann integrable} \iff D_f \text{ have zero measure}$$

*Proof.* We first prove:

$$f \text{ is proper Riemann integrable} \implies |D_f| = 0$$

Because  $D(\epsilon) \nearrow D_f$  as  $\epsilon \rightarrow 0$ , we must prove  $|D(\epsilon)| = 0$  for all  $\epsilon$ . Fix  $\epsilon$ . Because  $f$  is proper Riemann integrable, we know for all  $n$  there exists a partition  $P_n$  of  $[a, b]$  such that

$$U(P_n) - L(P_n) \leq \frac{1}{n}$$

Define  $\mathcal{G}_{n,\epsilon} \triangleq$  to be the union of closed intervals of  $P_n$  that have oscillation  $\geq \epsilon$ . By definition of pointwise oscillation, clearly we have

$$D(\epsilon) - P_n \subseteq \mathcal{G}_{n,\epsilon}$$

The proof now follows from the inequality:

$$\epsilon |D(\epsilon)| \leq \epsilon |\mathcal{G}_{n,\epsilon}| \leq \sum_{\text{osc}(f,I) \geq \epsilon} \text{osc}(f, I) |I| \leq U(P_n) - L(P_n)$$

which holds for all  $n$ . (done)

We now prove:

$$|D_f| = 0 \implies f \text{ is proper Riemann integrable}$$

Fix  $\epsilon$ . Let  $M \triangleq \sup |f|$ . Because  $D_f$  is measurable and null, there exists some  $G \supseteq D_f$  open in  $\mathbb{R}$  with  $|G| \leq \epsilon/4M$ .

Set  $K \triangleq [a, b] - G$ . Because for all  $x \in K$ ,  $f : [a, b] \rightarrow \mathbb{R}$  is continuous at  $x$ , we may select some closed interval  $I_x \subseteq [a, b]$  containing  $x$  such that  $\text{osc}(f, I_x) \leq \epsilon/2(b-a)$ . Moreover, because  $K$  is compact, we may write

$$K \subseteq I_{x_1} \cup \cdots \cup I_{x_n}$$

Consider the partition  $P \triangleq \{a, b\} \cup \partial I_{x_1} \cup \cdots \cup \partial I_{x_n}$ . Let  $J$  be a closed interval of  $P$ . Taking the fact that  $I_{x_1} \cup \cdots \cup I_{x_n}$  by nature only have finite number of connected components into account, we know  $J^\circ$  either completely lies in the union  $I_{x_1} \cup \cdots \cup I_{x_n}$  or completely lies outside of it, i.e., exactly one of the two statement:

$$J \subseteq I_{x_1} \cup \cdots \cup I_{x_n} \text{ or } J^\circ \subseteq [a, b] - I_{x_1} \cup \cdots \cup I_{x_n}$$

occurs. Moreover, if the former holds, we know  $J \subseteq I_{x_i}$  for some  $i$ , since if some  $y \in J^\circ$  lies in  $I_{x_j}$  for some  $j$ , then  $J \subseteq I_{x_j}$  by construction of  $P$ . Because of such, we may finally give

the desired estimate:

$$\begin{aligned} U(P) - L(P) &= \sum_{J \subseteq \tilde{I}} \text{osc}(f, J) |J| + \sum_{J^\circ \subseteq [a,b] - \tilde{I}} \text{osc}(f, J) |J| \\ &\leq \sum_{J \subseteq \tilde{I}} \frac{\epsilon}{2(b-a)} |J| + 2M \cdot \frac{\epsilon}{4M} \leq \epsilon \end{aligned}$$

where  $\tilde{I} \triangleq I_{x_1} \cup \dots \cup I_{x_n}$ . (done)

■

## 1.11 Miscellaneous

**Theorem 1.11.1. (Differentiability Theorem)** Suppose  $\alpha = \{e_1, \dots, e_n\}$  is an orthonormal basis of  $\mathbb{R}^n$ , and  $\beta = \{q_1, \dots, q_m\}$  is an orthonormal basis of  $\mathbb{R}^m$ . Suppose  $f$  maps an open set  $E \subseteq \mathbb{R}^n$  to  $\mathbb{R}^m$ . Then

$f$  is continuously differentiable on  $E \iff \partial_i f_j$  exists and is continuous on  $E$  for all  $i, j$

*Proof.* Skipped. ■

**Theorem 1.11.2. (Stirling's approximation)**

$$n! \sim \sqrt{2\pi n} \left(\frac{n}{e}\right)^n, \quad \text{as } n \rightarrow \infty$$

In other words, their ratio tends to 1 as  $n$  approaches  $\infty$ .

*Proof.* Skipped. ■

**Theorem 1.11.3. (Lagrange multiplier theorem)** Let  $V \subseteq \mathbb{R}^n$  be open,  $f, g_1, \dots, g_m \in C^1(V)$  with  $m \leq n$ , and  $\mathbf{y} \in V$ . If:

- (i)  $\mathbf{y}$  is a local extremum of  $f$  subject to the constraints  $g_1(\mathbf{x}) = \dots = g_m(\mathbf{x}) = 0$ .
- (ii)

$$\text{rank} \begin{pmatrix} \frac{\partial g_1}{\partial x_1}(\mathbf{y}) & \dots & \frac{\partial g_1}{\partial x_n}(\mathbf{y}) \\ \vdots & \ddots & \vdots \\ \frac{\partial g_m}{\partial x_1}(\mathbf{y}) & \dots & \frac{\partial g_m}{\partial x_n}(\mathbf{y}) \end{pmatrix} = m$$

then there exists a unique  $(\lambda_1, \dots, \lambda_m) \in \mathbb{R}^m$  such that

$$\nabla f(\mathbf{y}) = \sum_{i=1}^m \lambda_i \nabla g_i(\mathbf{y})$$

*Proof.* Skipped. ■

**Theorem 1.11.4. (Divergence theorem)** Let  $M \subseteq \mathbb{R}^3$  be compact with piecewise smooth boundary and  $\mathbf{F} : M \rightarrow \mathbb{R}^3$  be  $C^1$ . Then:

$$\int_{\partial M} \mathbf{F} \cdot \mathbf{n} dA = \int_M \nabla \cdot \mathbf{F} dV$$

*Proof.* This is **Stokes' theorem** applied to a 3-manifold embedded in  $\mathbb{R}^3$ . ■



**Theorem 1.11.5. (Green's theorem)** Let  $C \subseteq \mathbb{R}^2$  be a positively oriented, piecewise smooth, and simple closed curve. Let  $D \subseteq \mathbb{R}^2$  be the region bounded by  $C$ . If  $L, M \in C^1(\overline{D})$ , then

$$\int_C Ldx + Mdy = \int_D \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} \\ L & M \end{vmatrix} dA$$

*Proof.* This is **Stokes' theorem** applied to a 2-manifold embedded in  $\mathbb{R}^2$ . ■

**Theorem 1.11.6. (Multivariable Taylor's theorem)** Let  $U$  be a neighborhood of  $\mathbf{a} \in \mathbb{R}^n$ , and let  $f \in C^k(U)$ . Consider the multi-index notation

$$|\alpha| \triangleq \alpha_1 + \cdots + \alpha_n, \quad \alpha! \triangleq \alpha_1! \cdots \alpha_n!, \quad \mathbf{x}^\alpha \triangleq x_1^{\alpha_1} \cdots x_n^{\alpha_n}$$

We have:

$$f(\mathbf{x}) = f(\mathbf{a}) + \sum_{1 \leq |\alpha| \leq k} \frac{1}{\alpha!} \cdot \frac{\partial^{|\alpha|} f}{\partial x_1^{\alpha_1} \cdots \partial x_n^{\alpha_n}} \Big|_{\mathbf{a}} (\mathbf{x} - \mathbf{a})^\alpha + o(|\mathbf{x} - \mathbf{a}|^k)$$

*Proof.* Skipped. ■

# Chapter 2

## NTU Math M.A. Program Entrance Exam

### 2.1 Year 114

#### Question 1: easy estimate

Let  $\mathbf{x} = (x_1, x_2, x_3)$  be coordinate of  $\mathbb{R}^3$ . Let

$$f(\mathbf{x}) \triangleq \frac{x_1^3 + x_2^2 + i \ln(1 + x_2^2 + x_3^2)}{x_1^2 + x_2^2 + i x_3^2 \sin\left(\frac{x_1}{x_3}\right)} \quad \text{if } x_1 x_2 x_3 \neq 0$$
$$f(\mathbf{x}) \triangleq 0 \quad \text{if } x_1 x_2 x_3 = 0$$

Prove that

$$\{\mathbf{x} \in \mathbb{R}^3 : f \text{ is continuous at } \mathbf{x}\} = \{\mathbf{x} \in \mathbb{R}^3 : x_1 x_2 x_3 \neq 0\}$$

*Proof.* Because  $f$  is the quotient of two nonzero continuous function on  $\{\mathbf{x} \in \mathbb{R}^3 : x_1 x_2 x_3 \neq 0\}$ , we know  $f$  is continuous on it. Let

- (i)  $E_0 \triangleq \{\mathbf{0}\}$ .
- (ii)  $E_1 \triangleq \{\mathbf{x} \in \mathbb{R}^3 : x_1 = x_2 = 0\} - E_0$ .
- (iii)  $E_2 \triangleq \{\mathbf{x} \in \mathbb{R}^3 : x_1 = x_3 = 0\} - E_0$ .
- (iv)  $E_3 \triangleq \{\mathbf{x} \in \mathbb{R}^3 : x_2 = x_3 = 0\} - E_0$ .
- (v)  $E_4 \triangleq \{\mathbf{x} \in \mathbb{R}^3 : x_1 = 0\} - (E_0 \cup E_1 \cup E_2)$ .

(vi)  $E_5 \triangleq \{\mathbf{x} \in \mathbb{R}^3 : x_2 = 0\} - (E_0 \cup E_1 \cup E_3)$ .

(vii)  $E_6 \triangleq \{\mathbf{x} \in \mathbb{R}^3 : x_3 = 0\} - (E_0 \cup E_2 \cup E_3)$ .

We shall prove in order that all of them is contained by  $\{\mathbf{x} \in \mathbb{R}^3 : f \text{ is discontinuous at } \mathbf{x}\}$ .  
To see  $f$  is discontinuous at  $\mathbf{0}$ , just observe

$$\lim_{t \rightarrow 0} \operatorname{Re} f(\pi t, t, t) = \frac{1}{\pi^2 + 1}$$

Fix  $(0, 0, a) \in E_1$ . To see  $f$  is discontinuous at  $(0, 0, a)$ , just observe

$$|f(t, t, a + t)| = \frac{|t^3 + t^2 + i \ln(1 + t^2 + (a + t)^2)|}{\left|2t^2 + i(a + t)^2 \sin\left(\frac{t}{a+t}\right)\right|} \rightarrow \infty, \quad \text{as } t \rightarrow 0$$

Fix  $(0, a, 0) \in E_2$ . To see  $f$  is discontinuous at  $(0, a, 0)$ , just observe

$$\lim_{t \rightarrow 0} f(t, a + t, t) = \frac{a^2 + i \ln(1 + a^2)}{a^2}$$

Fix  $(a, 0, 0) \in E_3$ . To see  $f$  is discontinuous at  $(a, 0, 0)$ , just observe

$$\lim_{t \rightarrow 0} f(a + t, t, t) = a$$

Fix  $(0, a, b) \in E_4$ . To see  $f$  is discontinuous at  $(0, a, b)$ , just observe

$$\lim_{t \rightarrow 0} f(t, a + t, b + t) = \frac{a^2 + i \ln(1 + a^2 + b^2)}{a^2}$$

Fix  $(a, 0, b) \in E_5$ . To see  $f$  is discontinuous at  $(a, 0, b)$ , just observe

$$\lim_{t \rightarrow 0} f(a + t, b + t, 0) = \frac{a^3 + i \ln(1 + b^2)}{a^2 + ib^2 \sin\left(\frac{a}{b}\right)}$$

Fix  $(a, b, 0) \in E_6$ . To see  $f$  is discontinuous at  $(a, b, 0)$ , just observe

$$\lim_{t \rightarrow 0} f(a + t, t, b + t) = \frac{a^3 + b^2 + i \ln(1 + b^2)}{a^2 + b^2}$$

■

### Question 2: easy estimate

Let  $\mathbf{x} = (x_1, x_2, x_3)$  be coordinate of  $\mathbb{R}^3$ . Let

$$f(\mathbf{x}) \triangleq \frac{x_1^3 + e^{-\frac{1}{x^2}} + i \sin^2(x_3^2)}{x_1^2 + x_2^2 + ix_3^3} \quad \text{if } x_1x_2x_3 \neq 0$$

$$f(\mathbf{x}) \triangleq 0 \quad \text{if } x_1x_2x_3 = 0$$

Is  $f$  differentiable at  $\mathbf{0}$ ? (It isn't, as you may have guessed from the fact this is an exam question)

*Proof.* Because  $\frac{\partial f}{\partial x_i}|_{\mathbf{0}} = 0$  for all  $i$ , if  $f$  is differentiable at origin, then we must have  $df|_{\mathbf{0}} = (0, 0, 0)$ . In other words, we only have to check whether

$$\frac{|f(\mathbf{x}) - f(\mathbf{0})|}{|\mathbf{x}|} = \frac{|x_1^3 + e^{-\frac{1}{x^2}} + i \sin^2(x_3^2)|}{|x_1^2 + x_2^2 + ix_3^3| \sqrt{x_1^2 + x_2^2 + x_3^2}}$$

converge to 0 as  $\mathbf{x} \rightarrow \mathbf{0}$ . We claim it doesn't. Let  $\mathbf{x} = (t, t, t)$ . We have

$$\frac{|f(\mathbf{x}) - f(\mathbf{0})|}{|\mathbf{x}|} \geq \frac{|t^3|}{|2\sqrt{3}t^3 + \sqrt{3}it^4|} = \frac{1}{|2\sqrt{3} + \sqrt{3}it|} \rightarrow \frac{1}{2\sqrt{3}}, \quad \text{as } t \searrow 0$$

■

### Question 3: Inverse and implicit function theorem. Legendre transformation.

Let  $f(\mathbf{x}, \mathbf{y}) \in \mathcal{C}^\infty(\mathbb{R}^n \times \mathbb{R}^m)$ , where  $\mathbf{x} = (x_1, \dots, x_n)$  denote coordinates in  $\mathbb{R}^n$  and  $\mathbf{y} = (y_1, \dots, y_m)$  denote coordinates in  $\mathbb{R}^m$ . Suppose  $f(\mathbf{0}, \mathbf{0}) = 0$ ,  $\frac{\partial f}{\partial x_j}|_{(\mathbf{0}, \mathbf{0})} = 0$  for all  $x_j$ , and that the  $n$ -by- $n$  matrix  $\left(\frac{\partial^2 f}{\partial x_i \partial x_j}\right)|_{(\mathbf{0}, \mathbf{0})}$  is invertible.

(I) Show that there exists open neighborhood  $V$  of  $\mathbf{0} \in \mathbb{R}^m$  and smooth  $g : V \rightarrow \mathbb{R}^n$  such that

$$\frac{\partial f}{\partial x_j} \Big|_{(g(\mathbf{y}), \mathbf{y})} = 0, \quad \text{for all } \mathbf{y} \in V \text{ and } 1 \leq j \leq n$$

(II) Show that there exists open neighborhoods  $\Omega_1, \Omega \ni \mathbf{0} \in \mathbb{R}^n$  together with a

smooth function  $H : \Omega_1 \rightarrow \mathbb{R}$  that makes:

$$\left\{ \left( \mathbf{x}, \frac{\partial f}{\partial x_1} \Big|_{(\mathbf{x}, \mathbf{0})}, \dots, \frac{\partial f}{\partial x_n} \Big|_{(\mathbf{x}, \mathbf{0})} \right) \in \mathbb{R}^{2n} : \mathbf{x} \in \Omega \right\} \quad (2.1)$$

$$= \left\{ \left( \frac{\partial H}{\partial \xi_1} \Big|_{\boldsymbol{\xi}}, \dots, \frac{\partial H}{\partial \xi_n} \Big|_{\boldsymbol{\xi}}, \boldsymbol{\xi} \right) \in \mathbb{R}^{2n} : \boldsymbol{\xi} \in \Omega_1 \right\} \quad (2.2)$$

*Proof.* Define smooth  $\mathbf{h} : \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}^n$  by

$$\mathbf{h}(\mathbf{x}, \mathbf{y}) \triangleq \left( \frac{\partial f}{\partial x_1} \Big|_{(\mathbf{x}, \mathbf{y})}, \dots, \frac{\partial f}{\partial x_n} \Big|_{(\mathbf{x}, \mathbf{y})} \right)$$

(I) now follows from applying **implicit function theorem** to  $\mathbf{h}$ . For (II), first note that  $\mathbf{G} : \mathbb{R}^n \rightarrow \mathbb{R}^n$  defined by

$$\mathbf{G}(\mathbf{x}) \triangleq \left( \frac{\partial f}{\partial x_1} \Big|_{(\mathbf{x}, \mathbf{0})}, \dots, \frac{\partial f}{\partial x_n} \Big|_{(\mathbf{x}, \mathbf{0})} \right)$$

is smooth with full rank Jacobian at  $\mathbf{0} \in \mathbb{R}^n$ . This by **inverse function theorem** implies the existence of some neighborhood  $\Omega \ni \mathbf{0} \in \mathbb{R}^n$  such that  $\mathbf{G}$  forms a smooth diffeomorphism between  $\Omega$  and the open set  $\Omega_1 \triangleq \mathbf{G}(\Omega)$ . Because **set 2.1** is the graph of  $\mathbf{G}$  on  $\Omega$ , clearly we only have to show the existence of an  $H : \Omega_1 \rightarrow \mathbb{R}$  whose gradient is exactly  $\mathbf{G}^{-1} : \Omega_1 \rightarrow \Omega$ .<sup>1</sup>

Denote  $\mathbf{G}^{-1}$  by  $\mathbf{T} : \Omega_1 \rightarrow \Omega$ . We finish the proof by showing it suffices to define  $H : \Omega_1 \rightarrow \mathbb{R}$  by

$$H(\boldsymbol{\xi}) \triangleq \left( \sum_{i=1}^n \xi_i \cdot T_i(\boldsymbol{\xi}) \right) - f(\mathbf{T}(\boldsymbol{\xi}), \mathbf{0})$$

---

<sup>1</sup>Such is possible exactly because  $\mathbf{G}$  itself is a gradient.

Fix  $i$ . Compute

$$\begin{aligned}
\left. \frac{\partial H}{\partial \xi_i} \right|_{\xi} &= T_i(\xi) + \left( \sum_{j=1}^n \xi_j \cdot \left. \frac{\partial T_j}{\partial \xi_i} \right|_{\xi} \right) - \sum_{j=1}^n \left. \frac{\partial f}{\partial x_j} \right|_{(\mathbf{T}(\xi), \mathbf{0})} \cdot \left. \frac{\partial T_j}{\partial \xi_i} \right|_{\xi} \\
&= T_i(\xi) + \left( \sum_{j=1}^n \xi_j \cdot \left. \frac{\partial T_j}{\partial \xi_i} \right|_{\xi} \right) - \sum_{j=1}^n \mathbf{G}_j(\mathbf{T}(\xi)) \cdot \left. \frac{\partial T_j}{\partial \xi_i} \right|_{\xi} \\
&= T_i(\xi) + \left( \sum_{j=1}^n \xi_j \cdot \left. \frac{\partial T_j}{\partial \xi_i} \right|_{\xi} \right) - \sum_{j=1}^n \xi_j \cdot \left. \frac{\partial T_j}{\partial \xi_i} \right|_{\xi} = T_i(\xi)
\end{aligned}$$

■

#### Question 4: Convolution

Define for all  $k \in \mathbb{N}$  the function  $f_k : \mathbb{R}^n \rightarrow \mathbb{C}$  by:

$$f_k(\mathbf{x}) \triangleq k^{\frac{n}{2}} \int_{\mathbb{R}^n} e^{-k(|\mathbf{x}-\mathbf{y}|^2 + |\mathbf{x}-\mathbf{y}|^4) + i \sum_{j=1}^n x_j^2 y_j} d\mathbf{y}$$

Find its limit, and show that the convergence is always uniform on compact set.

*Proof.* Define  $\mathbf{z} \triangleq \sqrt{k}(\mathbf{y} - \mathbf{x})$ . **Change of variables formula** give us:

$$f_k(\mathbf{x}) = \int_{\mathbb{R}^n} e^{-|\mathbf{z}|^2 - \frac{|\mathbf{z}|^4}{k} + i \sum_{j=1}^n x_j^2 \left( \frac{z_j}{\sqrt{k}} + x_j \right)} d\mathbf{z}$$

Because the integrands are dominated by  $e^{-|\mathbf{z}|^2} \in L(\mathbb{R}^n)$ , we know by **DCT** that:

$$\lim_{k \rightarrow \infty} \int_{\mathbb{R}^n} e^{-|\mathbf{z}|^2 - \frac{|\mathbf{z}|^4}{k} + i \sum_{j=1}^n x_j^2 \left( \frac{z_j}{\sqrt{k}} + x_j \right)} d\mathbf{z} = \int_{\mathbb{R}^n} e^{-|\mathbf{z}|^2 + i \sum_{j=1}^n x_j^3} d\mathbf{z}$$

Fix  $\epsilon$  and  $M$ , and let  $K' \triangleq \{\mathbf{x} \in \mathbb{R}^n : |x_j| \leq M \text{ for all } j\}$ . We are required to find  $N$  such that:

$$\int_{\mathbb{R}^n} e^{-|\mathbf{z}|^2} \left| 1 - e^{-\frac{|\mathbf{z}|^4}{k} + i \sum_{j=1}^n \frac{x_j^2 z_j}{\sqrt{k}}} \right| d\mathbf{z} \leq \epsilon, \quad \text{for all } k \geq N \text{ and } \mathbf{x} \in K'$$

Let  $P > 0$  satisfies:

$$\int_{\mathbb{R}^n - K} e^{-|\mathbf{z}|^2} d\mathbf{z} \leq \frac{\epsilon}{4}, \quad \text{where } K \triangleq \{\mathbf{z} \in \mathbb{R}^n : |z_j| \leq P \text{ for all } j\} \quad (2.3)$$

Because we have:

$$\left| 1 - e^{-\frac{|\mathbf{z}|^4}{k} + i \sum_{j=1}^n \frac{x_j^2 z_j}{\sqrt{k}}} \right| \leq 2, \quad \text{for all } \mathbf{x}, \mathbf{z} \in \mathbb{R}^n$$

By [bound 2.3](#), we have:

$$\int_{\mathbb{R}^n - K} e^{-|\mathbf{z}|^2} \left| 1 - e^{-\frac{|\mathbf{z}|^4}{k} + i \sum_{j=1}^n \frac{x_j^2 z_j}{\sqrt{k}}} \right| d\mathbf{z} \leq \frac{\epsilon}{2}, \quad \text{for all } \mathbf{x} \in \mathbb{R}^n$$

Therefore, it only remains to find  $N$  so that

$$\int_K e^{-|\mathbf{z}|^2} \left| 1 - e^{-\frac{|\mathbf{z}|^4}{k} + i \sum_{j=1}^n \frac{x_j^2 z_j}{\sqrt{k}}} \right| d\mathbf{z} \leq \frac{\epsilon}{2}, \quad \text{for all } k \geq N \quad (2.4)$$

Clearly, the function  $g_k : K' \times K \rightarrow \mathbb{C}$  defined by:

$$g_k(\mathbf{x}, \mathbf{z}) \triangleq -\frac{|\mathbf{z}|^4}{k} + i \sum_{j=1}^n \frac{x_j^2 z_j}{\sqrt{k}}$$

converges uniformly to 0, which implies the function  $h_k : K' \times K \rightarrow \mathbb{R}$  defined by

$$h_k(\mathbf{x}, \mathbf{z}) \triangleq \left| 1 - e^{-\frac{|\mathbf{z}|^4}{k} + i \sum_{j=1}^n \frac{x_j^2 z_j}{\sqrt{k}}} \right|$$

converge to 0 uniformly. This implies the existence of some  $N$  such that

$$h_k(\mathbf{x}, \mathbf{z}) \leq \frac{\epsilon}{2} \cdot \pi^{-\frac{n}{2}}, \quad \text{for all } k \geq N \text{ and } (\mathbf{x}, \mathbf{z}) \in K' \times K$$

We have found the desired  $N$  for [inequality 2.4](#) ■

### Question 5: Convolution

Define for all  $k \in \mathbb{N}$  the function  $f_k : \mathbb{R}^n \rightarrow \mathbb{C}$  by:

$$f_k(\mathbf{x}) \triangleq k^{\frac{n}{2}} \int_{|\mathbf{y}| < M} e^{-k(|\mathbf{x}-\mathbf{y}|^2 + |\mathbf{x}-\mathbf{y}|^4) + i \sum_{j=1}^n x_j^2 y_j} d\mathbf{y}$$

Find its limit, and show that the convergence is always uniform on compact set.

*Proof.* Is the question even correct? ■

### Question 6: Arzelà–Ascoli theorem

Let  $U \subseteq \mathbb{R}$  be open and  $f_n : U \rightarrow \mathbb{R}$  be smooth. Suppose that for all compact  $K \subseteq U$  and  $m \in \mathbb{N} \cup \{0\}$ , there exists a constant  $C_{K,m} > 0$  such that

$$\sup \left\{ \left| f_n^{(m)}(x) \right| : x \in K, n \in \mathbb{N} \right\} \leq C_{K,m}$$

Suppose  $f_k$  has a pointwise limit  $f : U \rightarrow \mathbb{R}$ . Show that

- (i)  $f$  is smooth.
- (ii) For all  $m \in \mathbb{N} \cup \{0\}$  and compact  $K \subseteq U$ , sequence  $f_n^{(m)}$  converges to  $f^{(m)}$  uniformly on  $K$ .

*Proof.* We prove that for all  $m \in \mathbb{N} \cup \{0\}$ , we have:

$$f \in C^m(U) \quad \text{and} \quad f_n^{(m)} \rightarrow f^{(m)} \text{ uniformly on } K \text{ for all compact } K \subseteq U \quad (2.5)$$

The proof is done via induction on  $m$ . Clearly, for the base case we only have to show the latter. Fix compact  $K \subseteq U$ . It follows from **MVT** that  $\{f_n|_K\}$  are pointwise equicontinuous. We may now apply **Arzelà–Ascoli theorem** to see that every subsequence of  $\{f_n|_K\}$  has a uniformly convergent subsequence. Assume for a contradiction that  $f_n|_K$  doesn't uniformly converge to  $f|_K$ . Then, there exists  $\epsilon$ , some subsequence  $f_{n_k}$ , and some sequence  $\{x_k \in K\}$  such that  $|f_{n_k}(x_k) - f(x_k)| \geq \epsilon$  for all  $k \in \mathbb{N}$ . Such subsequence  $\{f_{n_k}\}$  clearly have no uniform convergent subsequence, a contradiction. Therefore, we have proved the base case.

We now prove the inductive case. The inductive hypothesis is

$$f \in C^{(m-1)}(U) \quad \text{and} \quad f_n^{(m-i)} \rightarrow f^{(m-i)} \text{ uniformly on } K \text{ for all } i \text{ and all compact } K \subseteq U$$

Because for all  $x \in U$ , we may find  $[x - \epsilon, x + \epsilon] \subseteq U$ , so to prove the inductive case, we only have to show the latter of **equation 2.5**. Method as in proof for the base case applies to show such. ■



## 2.2 Year 113

### Question 1: metric topology

If every closed and bounded set of a metric space  $M$  is compact, does it follow that  $(M, d)$  is complete?

*Proof.* Yes. Let  $\{x_n\}$  be a Cauchy sequence in  $M$ . To prove  $\{x_n\}$  converge in  $M$ , one let  $E$  be the closure of  $\{x_n\}$ , and prove that  $E$  is indeed bounded. This by premise implies  $E$  is compact, which implies there exists some convergent subsequence  $x_{n_k} \rightarrow x \in E$ , since compactness is equivalent to sequential compactness for metric space. The proof then follows from proving  $x_n \rightarrow x$ . ■

### Question 2: convergent divergent, Stirling's formula

Determine the values of  $h$  for which the following series converges uniformly on  $I_h = \{x \in \mathbb{R} : |x| \leq h\}$ .

$$\sum_{n=1}^{\infty} \frac{(n!)^2 x^n}{(2n)!} \quad (2.6)$$

*Proof.* Defining  $c_n \triangleq (n!)^2 / (2n)!$ , we may write series 2.6 as  $\sum c_n x^n$ . Because

$$\frac{c_{n+1}}{c_n} = \frac{(n+1)^2}{(2n+2)(2n+1)} \text{ converges to } \frac{1}{4}$$

and because root test is stronger than ratio test, i.e., we have

$$\liminf_{n \rightarrow \infty} \left| \frac{c_{n+1}}{c_n} \right| \leq \liminf_{n \rightarrow \infty} \sqrt[n]{|c_n|} \leq \limsup_{n \rightarrow \infty} \sqrt[n]{|c_n|} \leq \limsup_{n \rightarrow \infty} \left| \frac{c_{n+1}}{c_n} \right|$$

we know  $\sqrt[n]{c_n} \rightarrow \frac{1}{4}$ . This together with Weierstrass  $M$ -test and Cauchy-Hadamard theorem implies series 2.6 converges uniformly on  $I_h$  at least for any  $h < 4$ . We claim the series doesn't converge at  $x = 4$ . To see such, just use Stirling's formula to observe  $(n!)^2 4^n / (2n)! \sim \sqrt{\pi n} \rightarrow \infty$ . ■

### Question 3: measure-theoretic Feymann's trick

Consider

$$F(x) \triangleq \int_0^{\infty} \frac{e^{-xt} - e^{-t}}{t} dt$$

on  $I \triangleq \{x \in \mathbb{R} : \frac{1}{2} \leq x \leq 2\}$ .

- (i) Show that  $F$  is defined on  $I$  and  $F$  is continuous on  $I$ .
- (ii) Show that

$$F'(x) = \int_0^\infty -e^{-xt} dt$$

- (iii) Evaluate  $F(x)$

*Proof.* Fix  $x \in [\frac{1}{2}, 2]$ , and define  $f(t) \triangleq (e^{-xt} - e^{-t})/t$ . We first show:

- (i)  $f$  is integrable as  $t \rightarrow 0$ .
- (ii)  $f$  is integrable as  $t \rightarrow \infty$ .

The former is easy, as applying **L'Hospital rule**, we see  $f$  moreover converges at  $0^+$ . For the latter, we have to observe that because  $x \geq \frac{1}{2}$ , we have:

$$|e^{-xt} - e^{-t}| \leq |e^{-xt}| + |e^{-t}| \leq e^{\frac{-t}{2}} + e^{-t} \leq 2e^{\frac{-t}{2}} \quad (2.7)$$

for any large  $t$ . The latter now follows from **Comparison test** and the fact  $e^{-ct}$  integrable as  $t \rightarrow \infty$  for positive  $c$ . We now prove (ii) using **measure-theoretic Feymann's trick**, and the continuity of  $F$  will follow. To perform Feymann's trick, we have to establish the existence of some  $L^1$  function  $\theta : (0, \infty) \rightarrow \mathbb{R}$  such that for almost every  $t \in (0, \infty)$ , we have:

$$\left| \frac{\partial f}{\partial x}(x, t) \right| \leq \theta(t), \quad \text{for all } x \in \left[ \frac{1}{2}, 2 \right]$$

Clearly,  $\theta(t) \triangleq \exp(-\frac{t}{2})$  suffices. Lastly, we compute  $F(x)$  on  $[1/2, 2]$  using **FTC**:

$$F(1) = 0 \quad \text{and} \quad F'(x) = \int_0^\infty -e^{-xt} dt = \frac{1}{-x} \implies F(x) = \ln\left(\frac{1}{x}\right)$$

■

#### Question 4: inverse function theorem, Legendre transform

Consider smooth  $f : \mathbb{R}^n \rightarrow \mathbb{R}$  whose **Hessian Determinant** is 2 everywhere. We denote its **gradient**  $\nabla f : \mathbb{R}^n \rightarrow \mathbb{R}^n$  by  $F$ .

- (i) Show that there exists neighborhood  $U \subseteq \mathbb{R}^n$  of origin such that the restriction of  $F$  forms a smooth diffeomorphism from  $U$  to the image of  $U$ . Note that the

image is guaranteed to be open by **invariance of domain** if we really prove that the action is injective on  $U$ .

- (ii) Denote the inverse map in part (i) by  $\boldsymbol{\xi}(\mathbf{y}) \triangleq (\xi_1(\mathbf{y}), \dots, \xi_n(\mathbf{y}))$ . For any  $\mathbf{y}$  in the image of  $U$ , define

$$f^*(\mathbf{y}) \triangleq -f(\boldsymbol{\xi}(\mathbf{y})) + \sum_{i=1}^n y_i \xi_i(\mathbf{y})$$

Compute the Hessian Determinant of  $f^*$ .

*Proof.* (i) is a direct consequence of **inverse function theorem** and the observation that  $dF = \text{Hess } f$ . Fix  $k$ , and compute:

$$\begin{aligned} \left. \frac{\partial f^*}{\partial y_k} \right|_{\mathbf{y}} &= - \left( \sum_{j=1}^n \left. \frac{\partial f}{\partial x_j} \right|_{\boldsymbol{\xi}(\mathbf{y})} \left. \frac{\partial \xi_j}{\partial y_k} \right|_{\mathbf{y}} \right) + \left( \sum_{i \neq k} y_i \left. \frac{\partial \xi_i}{\partial y_k} \right|_{\mathbf{y}} \right) + \xi_k(\mathbf{y}) + y_k \left. \frac{\partial \xi_k}{\partial y_k} \right|_{\mathbf{y}} \\ &= - \left( \sum_{j=1}^n y_j \left. \frac{\partial \xi_j}{\partial y_k} \right|_{\mathbf{y}} \right) + \left( \sum_{i \neq k} y_i \left. \frac{\partial \xi_i}{\partial y_k} \right|_{\mathbf{y}} \right) + \xi_k(\mathbf{y}) + y_k \left. \frac{\partial \xi_k}{\partial y_k} \right|_{\mathbf{y}} \\ &= \xi_k(\mathbf{y}) \end{aligned}$$

Note that the second equality hold true by definition of  $\boldsymbol{\xi}$ . We now see that the matrix

$$\text{Hess } f^* = d\boldsymbol{\xi} = (dF)^{-1} = (\text{Hess } f)^{-1}$$

always have determinant  $\frac{1}{2}$ . ■

### Question 5: estimate with proof of contradiction

Let  $C^1$  function  $f : [0, \infty) \rightarrow \mathbb{R}$  satisfies:

- (i)  $f(x) \geq 0$  and  $f'(x) \leq 1$  for all  $x \geq 0$ .
- (ii)  $\int_0^\infty f(x) dx$  converges.

Does  $f(x)$  converge as  $x \rightarrow \infty$ ?

*Proof.* Yes. We will moreover show that  $f(x)$  converge to 0 as  $x \rightarrow \infty$ . Assume for a contradiction that there exists some  $\epsilon$  and  $x_n \nearrow \infty$  such that  $f(x_n) \geq \epsilon$  for all  $n$  and WLOG,  $x_{n+1} - x_n \geq \epsilon$ . Fix  $n$ . Because  $f$  is  $C^1$  with derivative always smaller than 1, by

**MVT**, every  $x \in [x_n - \epsilon, x_n)$  must satisfies  $[f(x_n) - f(x)]/(x_n - x) \leq 1$ . In other words, every  $x \in [x_n - \epsilon, x_n)$  satisfies

$$\epsilon - (x_n - x) \leq f(x_n) - (x_n - x) \leq f(x)$$

We may now estimate

$$\int_{x_n - \epsilon}^{x_n} f(x) dx \geq \int_{x_n - \epsilon}^{x_n} \epsilon - (x_n - x) dx = \frac{\epsilon^2}{2}$$

This cause a contradiction:

$$\int_0^\infty f(x) dx \geq \sum_n \int_{x_n - \epsilon}^{x_n} f(x) dx = \infty$$

■

## 2.3 Year 112

### Question 1: Cauchy-Schwarz for integral

Let

$$M \triangleq \left\{ f : [0, \infty] \rightarrow [0, \infty) \mid \int_0^\infty f^2(x) dx \leq 1 \right\}$$

Evaluate

$$\sup_{f \in M} \int_0^\infty f(x) e^{-x} dx \quad (2.8)$$

*Proof.* Use **Cauchy-Schwarz for integral** to show:

$$\sup_{f \in M} \int_0^\infty f(x) e^{-x} dx \leq \|f\|_2 \cdot \|e^{-x}\|_2 \leq \frac{1}{\sqrt{2}}$$

And (again use it to) show the equality holds when  $f = \sqrt{2}e^{-x}$ . ■

### Question 2: metric topology

Let  $a \in A \subseteq \mathbb{R}^n$ . Let  $A$  be compact, and let  $(a_n) \subseteq A$  be a sequence such that all convergent subsequences of  $(a_n)$  converges to  $a$ .

- (i) Does  $(a_n)$  converge to  $a$ ?
- (ii) If we remove the hypothesis of  $A$  being compact, does  $(a_n)$  converge to  $a$ ?

*Proof.* For (i), yes. Assume not for a contradiction. There exists some  $\epsilon$  and subsequence  $(a_{n_k})$  such that every  $a_{n_k}$  is  $\epsilon$ -away from  $a$ . By **definition of compact metric space**, there exists some convergent subsequence of  $(a_{n_k})$ , while being a subsequence of  $(a_n)$ , converges to a point other than  $a$ , a contradiction. For (ii), no. Let  $A \triangleq [0, 1)$  and  $a_n$  be  $\frac{1}{n}$  if  $n$  is even and  $1 - \frac{1}{n}$  if  $n$  is odd. ■

### Question 3: metric topology, proof by contradiction

Let  $A$  be some compact metric space, and let continuous  $f : A \rightarrow A$  never maps two points strictly closer to each other. Show that  $f$  is onto.

*Proof.* Assume  $a_0 \in A - f(A)$  for a contradiction, and define  $a_n \triangleq f^n(a_0)$ . Because  $f$  is continuous and  $A$  is compact, the image  $f(A)$  is compact. This implies the existence of

some positive real  $r$  smaller than the distance between  $a_0$  and  $p$  for any  $p \in f(A)$ . Because  $a_n \in f(A)$  for all positive  $n$ , we know  $d(a_n, a_0) \geq r$  for every positive  $n$ . Note that by induction, we have:

$$d(a_n, a_m) = d(f(a_{n-1}), f(a_{m-1})) \geq d(a_{n-1}, a_{m-1}) \geq r, \quad \text{for all } 0 \leq n < m. \quad (2.9)$$

The fact that  $\{a_n\}$  is a sequence in  $f(A)$  that satisfies **inequality 2.9** contradicts to the **sequential compactness** of  $f(A)$ . ■

#### Question 4: estimates of integrals

Define a sequence of function  $\{f_n(x)\}$  on  $[0, 1]$  as:

$$f_n(x) \triangleq \begin{cases} 1 & \text{if } x = 0 \\ 1 & \text{if } x \in (\frac{2k}{2^n}, \frac{2k+1}{2^n}], k \in \{0, 1, \dots, 2^{n-1} - 1\} \\ -1 & \text{if } x \in (\frac{2k+1}{2^n}, \frac{2k+2}{2^n}], k \in \{0, 1, \dots, 2^{n-1} - 1\} \end{cases}$$

Let  $g$  be a continuous function. Prove or disprove that  $\lim_{n \rightarrow \infty} \int_0^1 f_n g dx$  always converge to 0.

**Remark.** The original question, as stated, lacks precision. It doesn't specify the domain of the function  $g$ . If  $g$  is only defined on  $(0, 1)$ , then the product  $f_n g$  may fail to be integrable on  $(0, 1)$ , making the expression  $\int_0^1 f_n g dx$  ill-defined—for instance, if  $g(x) = \frac{1}{x}$ .

I will just suppose  $g$  is continuous on  $[0, 1]$ .

*Proof.* Write

$$\int_0^1 f_n g dx = \sum_{k=0}^{2^{n-1}-1} \int_{\frac{2k}{2^n}}^{\frac{2k+1}{2^n}} g(x) - g(x + \frac{1}{2^n}) dx$$

Because **continuous function on a compact domain is uniformly continuous**, for all  $\epsilon$ , there exists some  $N$  such that

$$\left| g(x) - g(x + \frac{1}{2^n}) \right| \leq \epsilon, \quad \text{for all } n \geq N \text{ and } x \in [0, 1 - \frac{1}{2^n}]$$

We may now estimate:

$$\left| \int_0^1 f_n g dx \right| \leq \sum_{k=0}^{2^{n-1}-1} \int_{\frac{2k}{2^n}}^{\frac{2k+1}{2^n}} \left| g(x) - g(x + \frac{1}{2^n}) \right| dx \leq \sum_{k=0}^{2^{n-1}-1} \frac{\epsilon}{2^n} = \frac{\epsilon}{2}$$

for all  $n \geq N$ . ■

### Question 5: $L^2$ -norm

Let  $P_2$  be the set of real polynomial with degree no greater than 2, and let  $S \triangleq \{p \in P_2 : p(1) = 1\}$ . Define  $G : P_2 \rightarrow \mathbb{R}$  by

$$G(p) \triangleq \int_0^1 p^2(t) dt$$

Does  $G|_S$  have any extreme value? If it does, find them.

**Remark:** The original phrasing of the question is misleading.

*Proof.* Clearly, the function  $G : P_2 \rightarrow \mathbb{R}$  is the same as the square of the norm function induced by the inner product:

$$\langle f, g \rangle \triangleq \int_0^1 f(t)g(t)dt \quad \text{for all } f, g \in P_2$$

By Gram-Schmidt, we see that  $P_2$  have orthonormal basis:

$$\left\{ 1, 2\sqrt{3} \left( t - \frac{1}{2} \right), 6\sqrt{5} \left( t^2 - t + \frac{1}{6} \right) \right\}$$

Computation shows that, with respect to this basis,  $S$  is the plane  $x + \sqrt{3}y + \sqrt{5}z = 1$ . Therefore, we know  $G$ , the square of the norm function, when restricted to  $S$  indeed has a local extremum value. It has only one, and it is a global minimum  $\frac{1}{9}$  at

$$p(t) \triangleq \frac{1}{9} \cdot 1 + \frac{\sqrt{3}}{9} \cdot 2\sqrt{3} \left( t - \frac{1}{2} \right) + \frac{\sqrt{5}}{9} \cdot 6\sqrt{5} \left( t^2 - t + \frac{1}{6} \right)$$

■

### Question 6: Lebesgue's criteria for proper Riemann integrability

Let  $C \in \mathbb{R}^+$ , function  $f : [a, b] \rightarrow \mathbb{R}$  be proper Riemann integrable, and  $g : \mathbb{R} \rightarrow \mathbb{R}$  be  $C$ -Lipschitz continuous. Prove that  $g \circ f : [a, b] \rightarrow \mathbb{R}$  is proper Riemann integrable.

**Remark:** The original question write "Riemann integrable" instead of "proper Riemann integrable." The statement doesn't hold true if we include improper Riemann integrable  $f$ . For example, let  $[a, b] \triangleq [0, 1]$ ,  $f(x) \triangleq (\sin(1/x))/x$ , and  $g(x) \triangleq |x|$ .

*Proof.* There are essentially two way to "answer" this question. One is to construct an honest bound. Another is to cheat by quoting **Lebesgue's criteria for proper Riemann**

**integrability.** The honest bound proof for this question and the proof for Lebesgue's criteria for proper Riemann integrability are morally the same. To quote Lebesgue's criteria for proper Riemann integrability, one only have to prove that  $g \circ f$  is continuous at  $x \in [a, b]$  if  $f$  is continuous at  $x$ . ■



## 2.4 Year 110

We say a topological space  $X$  is **locally compact** if for all  $x \in X$ , there exists some compact  $K \subseteq X$  such that  $x \in K^\circ$ . We say a map  $f : X \rightarrow Y$  is **proper** if  $f$  is continuous and for all compact  $K \subseteq Y$ ,  $f^{-1}(K)$  is compact.

### Question 1: Inverse function theorem, Proper map

Show that  $C^1$  proper map  $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$  (proper as preimage of compact subspace is compact) whose Jacobian is globally full rank is surjective.

*Proof.* Because  $\mathbb{R}^n$  is connected, to prove  $f$  is surjective we only have to prove image of  $f$  is clopen in  $\mathbb{R}^n$ . The image of  $f$  is open in  $\mathbb{R}^n$  by **inverse function theorem**. The image of  $f$  is closed in  $\mathbb{R}^n$  is morally a topological result. In particular, we prove **theorem 2.4.1**, which finish the proof. ■

**Theorem 2.4.1. (Proper map is closed, if the target is locally compact Hausdorff)** Let  $f : X \rightarrow Y$  be proper with  $Y$  locally compact Hausdorff, then for all closed  $A \subseteq X$ ,  $f(A) \subseteq Y$  is closed.

*Proof.* Fix closed  $A \subseteq X$ . Let  $y \in Y - f(A)$ . We are required to find neighborhood of  $y$  disjoint from  $f(A)$ . Because  $Y$  is locally compact, there exists compact  $K \subseteq Y$  such that  $y \in K^\circ$ . Therefore,  $f^{-1}(K)$  is compact by properness of  $f$ . Now, because  $E \triangleq A \cap f^{-1}(K)$  is closed in  $f^{-1}(K)$ , a compact space, we know  $E$  is compact.

It now follows from continuity of  $f$  that  $f(E)$  is compact. Because  $Y$  is Hausdorff, this implies  $f(E) \subseteq Y$  is closed.

The rest of the proof then follows from checking that  $K^\circ - f(E)$  is a neighborhood of  $y$  disjoint from  $f(A)$ . ■

### Question 2: divergence theorem

Let  $\mathbf{F} : \mathbb{R}^3 \rightarrow \mathbb{R}^3$  be the identity map, and let  $M \triangleq \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 \leq z \leq \sqrt{2 - (x^2 + y^2)}\}$  be positively oriented, so the normal vector field point "outward". Compute:

$$\int_{\partial M} \mathbf{F} \cdot \mathbf{n} dA$$

*Proof.* By **divergence theorem**, we have

$$\int_{\partial M} \mathbf{F} \cdot \mathbf{n} dA = \int_M \nabla \cdot \mathbf{F} dV = \int_M 3 dV$$

To perform **change of variables**, we set:

$$x = r \cos \theta \quad \text{and} \quad y = r \sin \theta$$

We have Jacobian  $|J_T| = |r|$ , and because  $x^2 + y^2 = \sqrt{2 - (x^2 + y^2)} \implies r^2 = x^2 + y^2 = 1$ , we also have:

$$T^{-1}(M) = \left\{ \theta \in [0, 2\pi] \quad \text{and} \quad r \in [0, 1] \quad \text{and} \quad z \in [r^2, \sqrt{2 - r^2}] \right\}$$

We may now compute

$$\int_M 3 dV = \int_0^{2\pi} \int_0^1 \int_{r^2}^{\sqrt{2-r^2}} 3r dz dr d\theta$$

Because using **change of variables**, we have:

$$\int_0^1 r \sqrt{2 - r^2} dr = \int_1^2 \frac{\sqrt{u}}{2} du = \frac{2^{\frac{3}{2}} - 1}{3}, \quad \text{where we set } u = 2 - r^2$$

We then have:

$$\int_0^{2\pi} \int_0^1 \int_{r^2}^{\sqrt{2-r^2}} 3r dz dr d\theta = 6\pi \int_0^1 \int_{r^2}^{\sqrt{2-r^2}} 3r dz dr d\theta = \pi \left( \frac{-3}{2} + 2 \left( 2^{\frac{3}{2}} - 1 \right) \right)$$

■

### Question 3: metric topology

Let  $M$  be a metric space with countable elements. Prove or disprove that  $M$  is connected.

*Proof.* If  $M$  has only one element, then  $M$  is trivially connected. We now prove that if countable  $M$  has more than one element then  $M$  is disconnected. Let  $x \in M$ . Because  $\{d(y, x) \in \mathbb{R}^+ : y \in M\}$  is countable, there exists some  $r$  such that  $d(y, x) \neq r$  for all  $y$ . The proof then follows from showing  $B_r(x)$  is closed. ■

### Question 4: high school math

Define  $f : \mathbb{R} \rightarrow \mathbb{R}$  by  $f(x) \triangleq \frac{1}{4} + x - x^2$ . Let  $f^2$  denote  $f \circ f$ , and  $f^n$  similarly.

- (i) Show that  $\{f^n(0)\}$  converges, and find its limit.
- (ii) Find all  $x \in \mathbb{R}$  that make  $\{f^n(x)\}$  converge to  $\lim f^n(0)$ .

*Proof.* Completing the square:

$$f(x) = \frac{1}{4} + x - x^2 = -(x - \frac{1}{2})^2 + \frac{1}{2}$$

One can see:

$$f^n(x) = -(x - \frac{1}{2})^{2^n} + \frac{1}{2}$$

which implies:

$$f^n(0) \rightarrow \frac{1}{2}$$

and moreover implies:

$$f^n(x) \rightarrow \frac{1}{2} \iff x \in \left(-\frac{1}{2}, \frac{3}{2}\right)$$

■

### Question 5: Holder space is Banach

Let  $X$  consist of all real-valued function  $f$  on  $[0, 1]$  such that  $f(0) = 0$  and

$$\|f\| \triangleq \sup_{x \neq y} \frac{|f(x) - f(y)|}{|x - y|^{\frac{1}{3}}} \quad \text{is finite.}$$

Show that  $X$  forms a vector space,  $\|\cdot\|$  forms a norm on  $X$ , and  $X$  is complete with respect to this norm.

Remark: More precisely, we are required to show  $X$  forms a  $\mathbb{R}$ -vector space.

*Proof.* To prove  $X$  forms a normed  $\mathbb{R}$ -vector space, we are required to prove:

- (i)  $0 \in X$
- (ii)  $\|f\| = 0 \implies f = 0$
- (iii) For all  $r \in \mathbb{R}$  and  $f \in X$ , we have  $\|rf\| = |r| \cdot \|f\|$

(iv) For all  $f, g \in X$ , we have  $\|f + g\| \leq \|f\| + \|g\|$

They are all easy. It remains to prove  $(X, \|\cdot\|)$  is Banach. Let  $\{f_n\} \subseteq (X, \|\cdot\|)$  be a Cauchy sequence. Proving  $\{f_n\}$  converges in  $(X, \|\cdot\|)$  consist of three steps:

- (i) Showing that for all  $x \in [0, 1]$ ,  $\{f_n(x)\} \subseteq \mathbb{R}$  is Cauchy. In particular,  $f_n \rightarrow f$  pointwise for some  $f : [0, 1] \rightarrow \mathbb{R}$ .
- (ii) Showing that  $f \in X$ .
- (iii) Showing that  $\|f_n - f\| \rightarrow 0$ .

Fix  $x \in [0, 1]$ . Our first goal is to prove that  $\{f_n(x)\} \subseteq \mathbb{R}$  is a Cauchy sequence. Fix  $\epsilon$ . Because  $\{f_n\} \subseteq X$  is Cauchy, we know there exists some  $N$  such that:

$$\|f_n - f_k\| \leq \epsilon x^{\frac{-1}{3}}, \quad \text{for all } n, k \geq N$$

It then follows from  $f_n(0) = f_k(0) = 0$  that:

$$|f_n(x) - f_k(x)| \leq |(f_n - f_k)(x) - (f_n - f_k)(0)| \leq \|f_n - f_k\| x^{\frac{1}{3}} \leq \epsilon, \quad \text{for all } n, k \geq N$$

We have shown  $\{f_n(x)\}$  is Cauchy. It then follows from completeness of  $\mathbb{R}$  that  $f_n \rightarrow f$  for some  $f : [0, 1] \rightarrow \mathbb{R}$  pointwise. We shall now show  $f \in X$ . Fix  $x \neq y$ . Because for all  $n$ , the inequality:

$$\begin{aligned} \frac{|f(x) - f(y)|}{|x - y|^{\frac{1}{3}}} &\leq \frac{|(f - f_n)(x) - (f - f_n)(y)|}{|x - y|^{\frac{1}{3}}} + \frac{|f_n(x) - f_n(y)|}{|x - y|^{\frac{1}{3}}} \\ &\leq \frac{|f(x) - f_n(x)|}{|x - y|^{\frac{1}{3}}} + \frac{|f(y) - f_n(y)|}{|x - y|^{\frac{1}{3}}} + \frac{|f_n(x) - f_n(y)|}{|x - y|^{\frac{1}{3}}} \end{aligned}$$

holds true, we know:

$$\frac{|f(x) - f(y)|}{|x - y|^{\frac{1}{3}}} \leq \sup \|f_n\| < \infty$$

In other words,  $\|f\| \leq \sup \|f_n\| < \infty$ , as desired. Lastly, it remains to prove  $\|f_n - f\| \rightarrow 0$ . Fix  $n$  and  $x \neq y$ . Clearly, we have:

$$\begin{aligned} \frac{|(f_n - f)(x) - (f_n - f)(y)|}{|x - y|^{\frac{1}{3}}} &= \lim_{k \rightarrow \infty} \frac{|(f_n - f_{n+k})(x) - (f_n - f_{n+k})(y)|}{|x - y|^{\frac{1}{3}}} \\ &\leq \sup_k \|f_n - f_{n+k}\| \end{aligned}$$

This implies:

$$\|f_n - f\| \leq \sup_k \|f_n - f_{n+k}\|, \quad \text{for all } n$$

which implies

$$\lim_{n \rightarrow \infty} \|f_n - f\| \leq \lim_{n \rightarrow \infty} \sup_k \|f_n - f_{n+k}\| = 0$$

as desired. ■

### Question 6: standard, MVT, proof by contradiction

Let  $f : (a, b) \rightarrow \mathbb{R}$  be differentiable. Show that  $f' : (a, b) \rightarrow \mathbb{R}$  has no jump discontinuity.

*Proof.* Assume for a contradiction that there exists some  $x \in (a, b)$  that makes  $f'(x-) \neq f'(x)$ , without loss of generality. By **MVT**, for each  $n \gg 0$  there exists some  $t_n \in (x - \frac{1}{n}, x)$  such that:

$$\frac{f(x) - f(x - \frac{1}{n})}{\frac{1}{n}} = f'(t_n)$$

The contradiction follows from noting

$$\lim_{n \rightarrow \infty} f'(t_n) = f'(x-) \neq f'(x) = \lim_{n \rightarrow \infty} \frac{f(x) - f(x - \frac{1}{n})}{\frac{1}{n}}$$
■

## 2.5 Year 109

### Question 1: Uniform implies $L^2$ , Orthogonality

Does there exist real sequence  $\{a_n\}$  that makes  $\sum_{n=2}^{\infty} a_n \sin(nx)$  converges uniformly to  $\sin(x)$  on  $[0, \pi]$ ?

*Proof.* No. Using formula:

$$\sin(nx) \sin(mx) = \frac{\cos((n-m)x) - \cos((n+m)x)}{2}$$

it is to compute that  $\sin(x)$  has standard  $L^2$ -distance  $\geq \sqrt{\pi/2}$  with any partial sum  $a_2 \sin(2x) + \cdots + a_N \sin(Nx)$ . Then, since uniform convergence on domain of finite measure implies  $L^2$ -convergence, it is impossible for  $\sum a_n \sin(nx)$  to converge uniformly to  $\sin(x)$ . ■

### Question 2: standard

Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be uniformly continuous. Show that  $f$  must be bounded by a linear function, i.e., there exists constant  $A, B \in \mathbb{R}$  such that

$$|f(x)| \leq A + B|x|, \quad \text{for all } x \in \mathbb{R}$$

*Proof.* Because  $f : \mathbb{R} \rightarrow \mathbb{R}$  is uniformly continuous. We know there exists some  $\delta > 0$  such that for all  $x, y \in \mathbb{R}$ , we have:

$$|x - y| \leq \delta \implies |f(x) - f(y)| \leq 1 \quad (2.10)$$

We claim

$$A \triangleq |f(0)| + 1 \text{ and } B \triangleq \frac{1}{\delta} \text{ suffices.}$$

By [inequality 2.10](#), we know

$$\max\{|f(n\delta)|, |f(-n\delta)|\} \leq |f(0)| + n, \quad \text{for all } n \in \mathbb{N} \cup \{0\}$$

Fix  $x$ . Let  $n$  satisfies  $n\delta < |x| \leq (n+1)\delta$ . By [inequality 2.10](#), we then have

$$|f(x)| \leq \max\{|f(n\delta)|, |f(-n\delta)|\} + 1 \leq |f(0)| + n + 1 \leq A + B|x|$$

■

### Question 3

Let  $M, N$  be two metric spaces, and let  $f : M \rightarrow N$  be a function.

- (i) Given the hypothesis that for each convergent sequence  $\{x_n\} \subseteq M$ , the sequence  $\{f(x_n)\} \subseteq N$  also converges, is  $f$  continuous?
- (ii) Suppose the function  $g : M \rightarrow \mathbb{R}$  makes  $E \triangleq \{(x, g(x)) \in M \times \mathbb{R} : x \in M\}$  compact in the product topology. Is  $g$  continuous?

*Proof.* The answers for both questions are positive. Assume for a contradiction that  $f$  is discontinuous at some  $x \in M$ . This implies the existence of some  $\epsilon$  such that for all  $\delta$ , we have  $f(B_\delta(x)) \not\subseteq B_\epsilon(f(x))$ . For all  $n \in \mathbb{N}$ , fix  $x_n \in B_{\frac{1}{n}}(x)$  that satisfies  $f(x_n) \notin B_\epsilon(f(x))$ . We now see that the image of the convergent sequence  $(x_1, x, x_2, x, x_3, x, \dots)$  doesn't converge, a contradiction.

Before we prove the continuity of  $g$ , we need to first make three necessary topological remarks:

- (a)  $g(M)$  is compact because  $E$  is compact and the projection  $M \times \mathbb{R} \rightarrow \mathbb{R}$  is continuous by definition of product topology.
- (b) Product topology have the property that if  $y_n \rightarrow y$  in topological space  $Y$  and  $z_n \rightarrow z$  in topological space  $Z$ , then  $(y_n, z_n) \rightarrow (y, z)$  in  $Y \times Z$ .

We may now prove the continuity of  $g$  using the result from question (i). Suppose  $x_n \rightarrow x$  in  $M$ . We are required to prove  $g(x_n) \rightarrow g(x)$ . Because  $g(M)$  is compact, we only have to prove that every sequential limit of  $g(x_n)$  is  $g(x)$ . Let  $g(x_{n_k})$  converges to  $g(y)$ . We now see by property of product topology that  $(x_{n_k}, g(x_{n_k})) \rightarrow (x, g(y)) \in E$ . This implies  $g(y) = g(x)$ , as desired. ■

### Question 4: Holder condition with exponent $> 1$ implies zero derivative at origin

Let  $L : \mathbb{R}^3 \rightarrow \mathbb{R}^3$  be an invertible linear map, and let function  $g : \mathbb{R}^3 \rightarrow \mathbb{R}^3$  be  $C^1$  while satisfying:

$$\text{For some } \epsilon, |g(\mathbf{x})| \leq C |\mathbf{x}|^{1+\epsilon} \text{ for all } \mathbf{x} \in \mathbb{R}^3$$

Show that  $L + g : \mathbb{R}^3 \rightarrow \mathbb{R}^3$  is invertible near the origin.

*Proof.* Because of **pointwise inverse function theorem**, we only have to show

$$dg|_0 = 0$$

Assume for a contradiction that  $dg|_0 \neq 0$ . Therefore, there exists some unit  $\mathbf{x} \in \mathbb{R}^3$  such that  $\mathbf{v} \triangleq dg|_0(\mathbf{x})$  is nonzero. Clearly  $g(\mathbf{0}) = \mathbf{0}$ . Therefore,

$$\frac{|g(t\mathbf{x}) - g(\mathbf{0}) - dg|_0(t\mathbf{x})|}{|t|} \geq \frac{|t\mathbf{v}| - C|t|^{1+\epsilon}}{|t|} = |\mathbf{v}| - C|t|^\epsilon \rightarrow |\mathbf{v}|, \quad \text{as } t \rightarrow 0$$

a contradiction. ■

### Question 5: change of variable, multiple integral

Find the volume of the ellipsoid:

$$(x + 2y)^2 + (x - 2y + z)^2 + 3z^2 \leq 1$$

*Proof.* To perform **change of variables**, we set:

$$t = \sqrt{3}z \quad \text{and} \quad u = x + 2y \quad \text{and} \quad v = x - 2y + z$$

which linearly and injectively maps the unit ball to

$$M \triangleq \{(x, y, z) \in \mathbb{R}^3 : (x + 2y)^2 + (x - 2y + z)^2 + 3z^2 \leq 1\}$$

Because the Jacobian  $|J_T|$  is  $\left| \frac{-\sqrt{3}}{12} \right|$ , we see that

$$\int_M dV = \int_{B_1(\mathbf{0})} \frac{\sqrt{3}}{12} dV = \frac{\sqrt{3}}{12} \cdot \text{Vol}(B_1(\mathbf{0})) = \frac{\sqrt{3}}{12} \cdot \frac{4}{3}\pi = \frac{\sqrt{3}\pi}{9}$$
■

### Question 6: Green's theorem

Let  $C \subseteq \mathbb{R}^2$  be a positively oriented simple closed curve. Find the curve  $C$  that maximize the integral:

$$\int_C y^3 dx + (3x - x^3) dy \tag{2.11}$$

*Proof.* Let  $D$  be the region bounded by  $C$ . By **Green's theorem**, we have

$$\int_C y^3 dx + (3x - x^3) dy = \int_D \left( \frac{d(3x - x^3)}{dx} - \frac{dy^3}{dy} \right) dA = \int_D (3 - 3x^2 - 3y^2) dA$$

We now see that the curve that maximize **value 2.11** is the unit circle  $x^2 + y^2 = 1$ . ■



### Question 7

Let  $f : [0, 1] \rightarrow \mathbb{R}$  be continuous. Find the limit:

$$\lim_{n \rightarrow \infty} (n+1) \int_0^1 x^n f(x) dx$$

*Proof.* Using **change of variable formula** with  $x = 1 - \frac{y}{n}$ , we see

$$\int_0^1 (n+1)x^n f(x) dx = \int_0^n \frac{n+1}{n} \left(1 - \frac{y}{n}\right)^n f\left(1 - \frac{y}{n}\right) dy$$

Noting that  $\left(1 - \frac{y}{n}\right)^n$  is dominated by  $e^{-y}$ , we see that by dominated convergence theorem the integral converge to  $f(1)$ . ■

## 2.6 Year 108

### Question 1: Alternating series test and Comparison test

Does the series

$$\sum_{n=1}^{\infty} \frac{(-1)^n \ln(n+1)}{n}$$

converge? Does it absolutely converge?

*Proof.* The fact that it converges follows from alternating series test<sup>2</sup>. The fact that it doesn't absolutely converge follows from comparing it with  $\sum \frac{1}{n}$ . ■

### Question 2

Define  $\Omega \triangleq \{(x, y) \in \mathbb{R}^2 : 0 \leq x, y \leq 1\} - \{(1, 1)\}$ , and define  $f : \Omega \rightarrow \mathbb{R}$  by

$$f(x, y) \triangleq \frac{1}{(1 - xy)^2}$$

- (i) For any  $\kappa \in (0, 1)$ , let  $U_\kappa \triangleq \{(x, y) \in \mathbb{R}^2 : 0 \leq x, y \leq \kappa\}$ . Is  $f$  uniformly continuous on  $U_\kappa$ ?
- (ii) Is  $f$  uniformly continuous on  $\Omega$ ?

*Proof.* Because  $U_\kappa$  are compact and  $f : \Omega \rightarrow \mathbb{R}$  is continuous, we know  $f$  is uniformly continuous on any  $U_\kappa$ . To see  $f$  is not uniformly continuous on  $\Omega$ , just note that

$$f(t, 1) \nearrow \infty, \quad \text{as } t \nearrow 1$$

### Question 3

Define  $f_n : [0, 1] \rightarrow \mathbb{R}$  for all  $n \in \mathbb{N}$  by  $f_n(x) \triangleq nx^n(1 - x)$ .

- (i) Determine  $\lim_{n \rightarrow \infty} f_n(x)$  for all  $x \in [0, 1]$ .
- (ii) Is the convergence uniform on  $[0, 1]$ ?

<sup>2</sup>Alternating series test follows from noting the partial sum sequence of the suspected series is Cauchy

*Proof.* Clearly  $f_n(x) \rightarrow 0$  for all  $x \in [0, 1]$ . Compute:

$$f_n \left( \frac{n}{n+1} \right) = \left( \frac{n}{n+1} \right)^{n+1} \rightarrow e^{-1}$$

■

#### Question 4

Define

$$F(x) \triangleq \int_0^\infty \frac{1 - \cos(xt)}{t^2 e^t} dt$$

- (i) Can we switch the order of differentiation and integration to obtain formulas for  $F'(x)$  and  $F''(x)$ ?
- (ii) Find explicit formulas (Not an improper integral) for  $F'(x)$  and  $F''(x)$ .

*Proof.* Define

$$f(x, t) \triangleq \frac{1 - \cos(xt)}{t^2 e^t}$$

Note that by **L'Hospital rule**, we have:

$$\lim_{t \rightarrow 0} \frac{1 - \cos(xt)}{t^2 e^t} = \frac{x^2}{2}$$

Therefore, the integral of  $f$  with respect to  $t$  has no singularities at 0. To see  $f$  also has no singularities at  $\infty$ , just compare its integral with that of  $2e^{-t}$ . We have shown that  $F$  is well-defined on  $\mathbb{R}$ . Compute

$$\frac{\partial f}{\partial x} = \frac{\sin(xt)}{te^t}$$

Recall that to perform **measure-theoretic Feymann's trick**, we are only required find some  $L^1$  function  $g$  such that for almost all  $x \in \mathbb{R}$ , we have:

$$\left| \frac{\partial f}{\partial x}(x, t) \right| \leq g(t), \quad \text{for all } t \in (0, \infty) \quad (2.12)$$

Because differentiation is a local notion, here, our  $g$  only has to satisfies **condition 2.12** for all  $|x| \leq M$ . Under the hypothesis that  $|x| \leq M$ , clearly we have:

$$\left| \frac{\sin(xt)}{te^t} \right| = \left| \frac{\sin(xt)}{xt} \right| \cdot \left| \frac{x}{e^t} \right| \leq \frac{M}{e^t}, \quad \text{for all } t \leq M$$

Therefore, we can construct  $g$  as:

$$g(t) \triangleq \begin{cases} \frac{1}{te^t} & \text{if } t > M \\ \frac{M}{e^t} & \text{if } t \leq M \end{cases}$$

Because

$$\frac{\partial^2 f}{\partial x^2} = \frac{\cos(xt)}{e^t}$$

is bounded by  $L^1$  function  $e^{-t}$ , **measure-theoretic Feymann's trick** is clearly justified. Computing that

$$\begin{aligned} \frac{d}{dt} (\sin(xt)e^{-t}) &= x \cos(xt)e^{-t} - \sin(xt)e^{-t} \\ \frac{d}{dt} (\cos(xt)e^{-t}) &= -x \sin(xt)e^{-t} - \cos(xt)e^{-t} \end{aligned}$$

We see

$$F''(x) = \int_0^\infty \frac{\cos(xt)}{e^t} dt = \left. \frac{x \sin(xt)e^{-t} - \cos(xt)e^{-t}}{x^2 + 1} \right|_{t=0}^\infty = \frac{1}{x^2 + 1}$$

This implies

$$F'(x) = \arctan(x) + C, \quad \text{for some } C \in \mathbb{R}$$

Because

$$F'(0) = \int_0^\infty \frac{0}{te^t} dt = 0$$

We know  $C = 0$ . ■

### Question 5: Implicit function theorem

Consider the function  $F : \mathbb{R}^4 \rightarrow \mathbb{R}^2$  defined by

$$F(x, y, u, v) \triangleq \left( \int_{x-y^2}^{x^2+y} (e^{t^2} + u) dt, x^3 + v \right)$$

- (i) Prove that near  $(1, 1, 0, 0)$ , the two equations  $F(x, y, u, v) = (\int_0^2 e^{t^2} dt, 1)$  can be solved for  $u, v$  as  $C^1$  functions of  $x, y$ .
- (ii) Find the first order partial derivatives of the functions  $u(x, y)$  and  $v(x, y)$  in part (i).

*Proof.* Direct computation give us:

$$\begin{aligned}\frac{\partial F_1}{\partial x} &= 2x \left( e^{(x^2+y)^2} + u \right) - \left( e^{(x-y^2)^2} + u \right) \\ \frac{\partial F_1}{\partial y} &= \left( e^{(x^2+y)^2} + u \right) + 2y \left( e^{(x-y^2)^2} + u \right) \\ \frac{\partial F_1}{\partial u} &= \int_{x-y^2}^{x^2+y} dt = (x^2 + y) - (x - y^2) \quad (\because \text{Feymann's trick}) \\ \frac{\partial F_1}{\partial v} &= 0\end{aligned}$$

And give use:

$$\begin{aligned}\frac{\partial F_2}{\partial x} &= 3x^2 \quad \text{and} \quad \frac{\partial F_2}{\partial y} = 0 \\ \frac{\partial F_2}{\partial u} &= 0 \quad \text{and} \quad \frac{\partial F_2}{\partial v} = 1\end{aligned}$$

Because  $F$  is  $C^1$ , and the matrix

$$\begin{pmatrix} \left. \frac{\partial F_1}{\partial u} \right|_{(1,1,0,0)} & \left. \frac{\partial F_1}{\partial v} \right|_{(1,1,0,0)} \\ \left. \frac{\partial F_2}{\partial u} \right|_{(1,1,0,0)} & \left. \frac{\partial F_2}{\partial v} \right|_{(1,1,0,0)} \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$$

is invertible, by **implicit function theorem**, indeed we can solve for  $u, v$  as  $C^1$  functions of  $x, y$  near  $(1, 1, 0, 0)$ . Because

$$F \circ \mathbf{g}(x, y) = \text{const.}, \quad \text{where } \mathbf{g}(x, y) \triangleq (x, y, u(x, y), v(x, y))$$

We know

$$\begin{aligned}\begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{pmatrix} &= \begin{pmatrix} \frac{\partial F_1}{\partial u} & \frac{\partial F_1}{\partial v} \\ \frac{\partial F_2}{\partial u} & \frac{\partial F_2}{\partial v} \end{pmatrix}^{-1} \begin{pmatrix} -\begin{pmatrix} \frac{\partial F_1}{\partial x} & \frac{\partial F_1}{\partial y} \\ \frac{\partial F_2}{\partial x} & \frac{\partial F_2}{\partial y} \end{pmatrix} \\ \end{pmatrix} \\ &= \begin{pmatrix} \frac{-\left(2x\left(e^{(x^2+y)^2}+u(x,y)\right)-\left(e^{(x-y^2)^2}+u\right)\right)}{(x^2+y)-(x-y^2)} & \frac{-\left(\left(e^{(x^2+y)^2}+u\right)+2y\left(e^{(x-y^2)^2}+u\right)\right)}{(x^2+y)-(x-y^2)} \\ -3x^2 & 0 \end{pmatrix}\end{aligned}$$

■

## 2.7 Year 107

### Question 1

Evaluate the integral

$$\int_{\frac{\pi}{4}}^{\frac{5\pi}{4}} \frac{1}{1 + 3 \sin^2(x)} dx$$

Hint: Let  $u = \tan x$ .

*Proof.* Clearly, we have the symmetry:

$$\int_{\frac{\pi}{4}}^{\frac{5\pi}{4}} \frac{1}{1 + 3 \sin^2(x)} dx = 2 \int_0^{\frac{\pi}{2}} \frac{1}{1 + 3 \sin^2(x)} dx$$

Using **change of variable formula** with  $u = \tan x$ , we have

$$\begin{aligned} \int_0^{\frac{\pi}{2}} \frac{1}{1 + 3 \sin^2(x)} dx &= \int_0^{\frac{\pi}{2}} \frac{\sec^2(x)}{\sec^2(x) + 3 \tan^2(x)} dx \\ &= \int_0^{\frac{\pi}{2}} \frac{\sec^2(x)}{1 + 4 \tan^2(x)} dx \\ &= \int_0^{\infty} \frac{1}{1 + 4u^2} du = \frac{\arctan(2u)}{2} \Big|_{u=0}^{\infty} = \frac{\pi}{4} \end{aligned}$$

■

### Question 2

State and prove Leibniz criterion for convergence of alternating series.

*Proof.* **Alternating series test** stated that:

$$a_n \searrow 0 \implies \sum (-1)^n a_n \text{ converges.}$$

Fix  $\epsilon$ . We are required to find some  $N$  such that

$$\left| \sum_{k=N}^{N+n} (-1)^k a_k \right| \leq \epsilon, \quad \text{for all } n \geq 0$$

Because  $a_n \searrow 0$ , we know there exist some even  $N$  such that

$$\epsilon \geq a_N \geq a_{N+1} \geq \cdots \geq 0$$

We shall prove such  $N$  suffices. The base case  $n = 0$  is clear. Before we prove the inductive case, we shall make the remark that grouping the sum in the form:

$$\sum_{k=N}^{N+n} (-1)^k a_k = (a_N - a_{N+1}) + (a_{N+2} - a_{N+3}) + \cdots$$

where the last term is either  $(a_{N+n})$  or  $(a_{N+n-1} - a_{N+n})$  depending on parity of  $n$ , we see the sum  $\sum_{k=N}^{N+n} (-1)^k a_k$  is always nonnegative, since all terms are nonnegative.

We now prove the inductive case. Suppose the inequality hold true for all  $n \leq m$ . To prove the inductive case, we are required to prove the assertion also holds true for  $n = m + 1$ . Either  $n$  is even or odd. If  $n$  is odd, then the proof finish, since

$$0 \leq \sum_{k=N}^{N+n} (-1)^k a_k \leq \sum_{k=N}^{N+m} (-1)^k a_k \leq \epsilon$$

as desired. If  $n$  is even, then the proof also finish, since

$$\begin{aligned} 0 &\leq \sum_{k=N}^{N+n} (-1)^k a_k = \left( \sum_{k=N}^{N+n-2} (-1)^k a_k \right) + (-a_{N+n-1} + a_{N+n}) \\ &\leq \sum_{k=N}^{N+n-2} (-1)^k a_k \leq \epsilon \end{aligned}$$

■

### Question 3: Proof of second partial derivative test

Let  $\mathbf{p} \in \mathbb{R}^2$  and let smooth  $F : \mathbb{R}^2 \rightarrow \mathbb{R}$  satisfies:

$$\begin{aligned} \frac{\partial F}{\partial x} \Big|_{\mathbf{p}} &= \frac{\partial F}{\partial y} \Big|_{\mathbf{p}} = 0 \\ \frac{\partial^2 F}{\partial x^2} \Big|_{\mathbf{p}} &> 0 \\ \left( \frac{\partial^2 F}{\partial x^2} \cdot \frac{\partial^2 F}{\partial y^2} - \left( \frac{\partial^2 F}{\partial x \partial y} \right)^2 \right) \Big|_{\mathbf{p}} &> 0 \end{aligned}$$

Show that  $\mathbf{p}$  is a local minimum of  $F$ .

*Proof.* Recall that the Hessian matrices  $\text{Hess } F$  are symmetric, so by spectral theory for finite dimensional real vector space, there exists some  $P \in \text{O}_2(\mathbb{R})$  such that

$$P(\text{Hess } F)|_{\mathbf{p}}P^t = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}, \quad \text{for some } \lambda_1, \lambda_2 \in \mathbb{R}$$

The premise said that the determinant of the Hessian  $(\text{Hess } F)|_{\mathbf{p}}$  is positive, so we know  $\lambda_1, \lambda_2$  are both nonzero and share the same sign. Because

$$\mathbf{h}^t(\text{Hess } F)|_{\mathbf{p}}\mathbf{h} = (P\mathbf{h})^t (P(\text{Hess } F)|_{\mathbf{p}}P^t) (P\mathbf{h}) = \lambda_1(P\mathbf{h})_1^2 + \lambda_2(P\mathbf{h})_2^2$$

and because

$$\begin{pmatrix} 1 & 0 \end{pmatrix} \cdot (\text{Hess } F)|_{\mathbf{p}} \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{\partial^2 F}{\partial x^2} \Big|_{\mathbf{p}} > 0$$

we know that  $\lambda_1, \lambda_2$  are both positive. Because  $F$  is smooth, and  $\nabla F|_{\mathbf{p}} = 0$  by premise, we may use **multivariables Taylor's theorem** to deduce:

$$\begin{aligned} F(\mathbf{p} + \mathbf{h}) &= F(\mathbf{p}) + \nabla F|_{\mathbf{p}} \cdot \mathbf{h} + \frac{1}{2} \cdot \mathbf{h}^t(\text{Hess } F|_{\mathbf{p}})\mathbf{h} + o(|\mathbf{h}|^2) \\ &= F(\mathbf{p}) + \frac{1}{2} \cdot \mathbf{h}^t(\text{Hess } F|_{\mathbf{p}})\mathbf{h} + o(|\mathbf{h}|^2) \\ &= F(\mathbf{p}) + \frac{1}{2} (\lambda_1 c_1^2 + \lambda_2 c_2^2) + o(c_1^2 + c_2^2), \quad \text{where } \mathbf{h} = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 \end{aligned}$$

where  $\mathbf{v}_1, \mathbf{v}_2$  are the unital eigenvector of  $\text{Hess } F|_{\mathbf{p}}$ . This shows that  $\mathbf{p}$  is indeed a local minimum. ■

#### Question 4: Arzelà–Ascoli theorem

Let  $X$  be a metric space. We say a sequence  $(f_n : X \rightarrow \mathbb{R})$  of real-valued functions **converges compactly** if for every compact  $K \subseteq X$ , the sequence  $(f_n|_K : K \rightarrow \mathbb{R})$  converges uniformly.

Show that if a pointwise equicontinuous sequence  $(f_n : X \rightarrow \mathbb{R})$  of real-valued functions pointwise converge to some continuous function  $f : X \rightarrow \mathbb{R}$ , then the convergence is compact.

*Proof.* Fix compact  $K \subseteq X$ . Assume for a contradiction that the convergence to  $f$  is not uniform on  $K$ . Therefore, there exists some  $\epsilon$  and some subsequence  $\{f_{n_k}(x_{n_k}) \in \mathbb{R} : x_{n_k} \in K\}$  such that

$$|f_{n_k}(x_{n_k}) - f(x_{n_k})| \geq \epsilon, \quad \text{for all } k \in \mathbb{N} \tag{2.13}$$



Because  $(f_n)$  pointwise converges, we know  $(f_n)$  is pointwise bounded. Therefore, by **Arzelà–Ascoli theorem**, the subsequence  $(f_{n_k}|_K)$  has a uniformly convergent subsequence, which is impossible by **condition 2.13**, a contradiction. ■

### Question 5: Divergence theorem and integration on tours

Compute the outward flux of the vector field  $\mathbf{F} \triangleq (x + ye^z, e^x \sin(yz), ye^{zx})$  through the boundary of the region  $M \triangleq \{(x, y, z) \in \mathbb{R}^3 : (\sqrt{x^2 + y^2} - 3)^2 + z^2 < 1\}$ .

*Proof.* Compute

$$(\nabla \cdot \mathbf{F})(x, y, z) = 1 + ze^x \cos(yz) + xye^{zx}$$

By **divergence theorem**, we have

$$\begin{aligned} \int_{\partial M} \mathbf{F} \cdot \mathbf{n} dA &= \int_M \nabla \cdot \mathbf{F} dV \\ &= \int_M (1 + ze^x \cos(yz) + xye^{zx}) dV \\ &= \int_M dV \end{aligned}$$

where the last equality hold true because  $ze^x \cos(yz)$  and  $xye^{zx}$  are skew-symmetric with respect to  $z$  and  $y$ , respectively. We may now use **change of variable formula** and **Fubini's theorem** to compute:

$$\begin{aligned} \int_M dV &= \int_{\widetilde{M}} r dr d\theta dz \\ &= 2\pi \int_{u^2 + z^2 \leq 1} (u + 3) dA = 6\pi^2 \end{aligned}$$

where  $\widetilde{M} = \{(r, \theta, z) \in \mathbb{R}^3 : (r - 3)^2 + z^2 < 1 \text{ and } 0 \leq \theta \leq 2\pi\}$  ■

### Question 6: Sequence and Series, series $\sum \cos(ne)$ is bounded

Show that the function

$$f(x) \triangleq \sum_{n=1}^{\infty} \frac{\cos(ne)}{n^x}$$

is well-defined and continuous on  $(0, \infty)$ .

*Proof.* We are required to show that for all  $x$ , there exist some  $B_\epsilon(x)$  on which the series uniformly converge. Because by **M-test**, the series converge uniformly on  $[2, \infty)$ , we only have to show such for  $x \in (0, 2]$ . This is done via **Dirichlet's test for uniform convergence**. Fix  $x_0 \in (0, 2]$ . We are required to show that:

- (i) The sequence  $\frac{1}{n^x}$  uniformly converges to 0 on  $B_\epsilon(x_0)$ .
- (ii) The series  $\sum \cos(ne)$  is (uniformly) bounded.

Fact (i) is clear. For (ii), simply observe

$$\sum_{n=1}^N \cos(ne) = \sum_{n=1}^N \frac{e^{ine} + e^{-ine}}{2} = \frac{1}{2} \left( \frac{e^{ie} - e^{i(N+1)e}}{1 - e^{ie}} + \frac{e^{-ie} - e^{-i(N+1)e}}{1 - e^{-ie}} \right)$$

■

### Question 7: Implicit function theorem

Let  $f, g : \mathbb{R}^2 \rightarrow \mathbb{R}$  be smooth, and define  $S \triangleq \{(x, y) \in \mathbb{R}^2 : f(x, y) = 0\}$ . Suppose  $\mathbf{p} \in S$  satisfies

$$\begin{aligned} \nabla f|_{\mathbf{p}} &= (-1, 2) \quad \text{and} \quad \nabla g|_{\mathbf{p}} = (3, -6) \\ (\text{Hess } f)|_{\mathbf{p}} &= \begin{pmatrix} 1 & 3 \\ 3 & 0 \end{pmatrix} \quad \text{and} \quad (\text{Hess } g)|_{\mathbf{p}} = \begin{pmatrix} 3 & -1 \\ -1 & 2 \end{pmatrix} \end{aligned}$$

Show that  $\mathbf{p}$  is a local minimum for  $g|_S$ .

*Proof.* By **implicit function theorem**, there exists some  $B_\epsilon(\mathbf{p}) \subseteq \mathbb{R}^2$  and smooth  $h : I \rightarrow \mathbb{R}$  such that

$$S \cap B_\epsilon(\mathbf{p}) = \{(x, h(x)) \in \mathbb{R}^2 : x \in I\} \quad \text{and} \quad \mathbf{p} = (t_0, h(t_0))$$

Define  $G : I \rightarrow \mathbb{R}$  by

$$G(t) \triangleq g(t, h(t))$$

By second derivative test, we only have to prove that:

$$G'(t_0) = 0 \quad \text{and} \quad G''(t_0) > 0$$

Because  $f(t, h(t))$  by definition vanishes identically on  $I$ , we know

$$\frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} \cdot h'(t) = 0$$

which implies

$$h'(t) = -\frac{\partial f}{\partial x} \cdot \left( \frac{\partial f}{\partial y} \right)^{-1}$$

and implies

$$h''(t) = -\frac{\left( \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial x \partial y} \cdot h'(t) \right) \frac{\partial f}{\partial y} - \frac{\partial f}{\partial x} \left( \frac{\partial^2 f}{\partial x \partial y} + \frac{\partial^2 f}{\partial y^2} \cdot h'(t) \right)}{\left( \frac{\partial f}{\partial y} \right)^2}$$

Compute

$$G'(t) = \frac{\partial g}{\partial x} + \frac{\partial g}{\partial y} \cdot h'(t)$$

Compute

$$G''(t) = \frac{\partial^2 g}{\partial x^2} + \frac{\partial^2 g}{\partial x \partial y} \cdot h'(t) + \left( \frac{\partial^2 g}{\partial x \partial y} + \frac{\partial^2 g}{\partial y^2} \cdot h'(t) \right) \cdot h'(t) + \frac{\partial g}{\partial y} \cdot h''(t)$$

We then see

$$G'(t_0) = 0 \text{ and } G''(t_0) = \frac{29}{2} > 0$$

■

## 2.8 Year 106

### Question 1: tricky multiple integral

Let  $M$  be the unit ball  $B_1(0)$  in  $\mathbb{R}^3$ . Compute

$$\int_M \cos(x + y + z) dx dy dz$$

*Proof.* Defining vectors  $\mathbf{v}, \mathbf{w}, \mathbf{s} \in \mathbb{R}^3$  by

$$\mathbf{v} \triangleq (1, 1, 1) \quad \text{and} \quad \mathbf{w} \triangleq \left( \frac{1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}, 0 \right) \quad \text{and} \quad \mathbf{s} \triangleq \left( \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{-2}{\sqrt{6}} \right)$$

We have an injective differentiable map:

$$\begin{aligned} (t, \theta, r) &\mapsto t\mathbf{v} + r(\cos \theta)\mathbf{w} + r(\sin \theta)\mathbf{s} \\ &= \left( t + \frac{r \cos \theta}{\sqrt{2}} + \frac{r \sin \theta}{\sqrt{6}}, t + \frac{-r \cos \theta}{\sqrt{2}} + \frac{r \sin \theta}{\sqrt{6}}, t + \frac{-2r \sin \theta}{\sqrt{6}} \right) \end{aligned}$$

Such that when we perform the **change of variable** with **Fubini's theorem**, we see:

$$\int_M \cos(x + y + z) dV = \int_0^{2\pi} \int_{\frac{-1}{\sqrt{3}}}^{\frac{1}{\sqrt{3}}} \int_0^{\sqrt{1-3t^2}} \cos(3t) \sqrt{3} r dr dt d\theta$$

with the Jacobian being:

$$\left| \det \begin{pmatrix} \frac{\partial x}{\partial t} & \frac{\partial y}{\partial t} & \frac{\partial z}{\partial t} \\ \frac{\partial x}{\partial r} & \frac{\partial y}{\partial r} & \frac{\partial z}{\partial r} \\ \frac{\partial x}{\partial \theta} & \frac{\partial y}{\partial \theta} & \frac{\partial z}{\partial \theta} \end{pmatrix} \right| = \sqrt{3}r$$

Compute

$$\int_0^{2\pi} \int_{\frac{-1}{\sqrt{3}}}^{\frac{1}{\sqrt{3}}} \int_0^{\sqrt{1-3t^2}} \cos(3t) \sqrt{3} r dr dt d\theta = \sqrt{3}\pi \int_{\frac{-1}{\sqrt{3}}}^{\frac{1}{\sqrt{3}}} \cos(3t)(1 - 3t^2) dt$$

Because

$$\cos(t)t^2 = \frac{d}{dt} ((\sin t)t^2 + 2(\cos t)t - 2\sin t)$$

We may compute

$$\sqrt{3}\pi \int_{\frac{-1}{\sqrt{3}}}^{\frac{1}{\sqrt{3}}} \cos(3t)(1 - 3t^2) dt = \pi \left[ \sin(\sqrt{3}) \left( \frac{4\sqrt{3}}{9} \right) + \cos(\sqrt{3}) \left( \frac{-4}{3} \right) \right]$$

■

## Question 2: Lebesgue's criteria, Bessel's inequality

Let  $f : [-\pi, \pi] \rightarrow \mathbb{R}$  be proper Riemann integrable, and define its Fourier coefficients by

$$\begin{aligned} a_n &\triangleq \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) dx, & \text{for all } n \in \mathbb{N} \cup \{0\} \\ b_n &\triangleq \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx) dx, & \text{for all } n \in \mathbb{N} \end{aligned}$$

- (i) Show that  $f^2 : [-\pi, \pi]$  is proper Riemann integrable.
- (ii) Show that the series  $\sum a_n^2 + b_n^2$  converges.

*Proof.* (i) is a direct consequence of the fact that if  $f : [-\pi, \pi] \rightarrow \mathbb{R}$  is continuous at  $x \in [-\pi, \pi]$ , then  $f^2 : [-\pi, \pi] \rightarrow \mathbb{R}$  is continuous at  $x$ , and **Lebesgue's criteria for proper Riemann integrability**.

(ii) is just Bessel's inequality.<sup>3</sup> Let  $N \geq 0$ . We have:

$$\begin{aligned} &\frac{1}{\pi} \int_{-\pi}^{\pi} \left( \sum_{n=0}^N a_n \cos(nx) + b_n \sin(nx) \right) \left( f(x) - \sum_{n=0}^N a_n \cos(nx) + b_n \sin(nx) \right) dx \\ &= \left( \sum_{n=0}^N a_n^2 + b_n^2 \right) - \frac{1}{\pi} \int_{-\pi}^{\pi} \left( \sum_{n=0}^N a_n \cos(nx) + b_n \sin(nx) \right)^2 dx \\ &= \left( \sum_{n=0}^N a_n^2 + b_n^2 \right) - \left( \sum_{n=0}^N a_n^2 + b_n^2 \right) = 0 \end{aligned}$$

This give us:

$$\begin{aligned} \int_{-\pi}^{\pi} f^2(x) dx &= \int_{-\pi}^{\pi} \left( \sum_{n=0}^N a_n \cos(nx) + b_n \sin(nx) \right)^2 dx \\ &\quad + \int_{-\pi}^{\pi} \left( f(x) - \sum_{n=0}^N a_n \cos(nx) + b_n \sin(nx) \right)^2 dx \\ &= \pi \left( \sum_{n=0}^N a_n^2 + b_n^2 \right) + \int_{-\pi}^{\pi} \left( f(x) - \sum_{n=0}^N a_n \cos(nx) + b_n \sin(nx) \right)^2 dx \end{aligned}$$

<sup>3</sup>I forgot to consider  $\int_{-\pi}^{\pi} 1 = 2\pi$ , but this mistake doesn't hurt the proof.

Which moreover give us:

$$\sum_{n=0}^N a_n^2 + b_n^2 \leq \frac{1}{\pi} \int_{-\pi}^{\pi} f^2(x)$$

It then follows from monotone convergence theorem that  $\sum a_n^2 + b_n^2$  converges. ■

### Question 3: Dirichlet's test

(i) Let  $\{a_n\}$  and  $\{b_n\}$  be two sequences of real numbers, and define  $B_n \triangleq b_1 + \cdots + b_n$ . Show that if  $a_n \searrow 0$  and  $\{B_n\}$  is bounded, then the series  $\sum a_n b_n$  converges.

(ii) Show that the function series:

$$\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} \sin(nx)$$

converges uniformly on  $[-\pi + \epsilon, \pi - \epsilon]$ .

*Proof.* (i) is just **Dirichlet's test**. Recall **Dirichlet's test for uniform convergence**. To prove (ii), we only have to prove that  $\sum (-1)^{n-1} \sin(nx)$  is uniformly bounded on  $[-\pi + \epsilon, \pi - \epsilon]$ . Because  $(-1)^{n-1} = -e^{in\pi}$ , we may compute

$$\begin{aligned} \sum_{n=1}^N (-1)^{n-1} \sin(nx) &= \sum_{n=1}^N (-1)^{n-1} \frac{e^{inx} - e^{-inx}}{2i} = \sum_{n=1}^N \frac{e^{in(\pi-x)} - e^{in(\pi+x)}}{2i} \\ &= \frac{1}{2i} \cdot \left( \frac{e^{i(\pi-x)} - e^{i(N+1)(\pi-x)}}{1 - e^{i(\pi-x)}} - \frac{e^{i(\pi+x)} - e^{i(N+1)(\pi+x)}}{1 - e^{i(\pi+x)}} \right) \end{aligned}$$

The rest then follows from noting both  $\pi + x$  and  $\pi - x$  must lie in  $[\epsilon, 2\pi - \epsilon]$ , which implies the series  $\sum (-1)^{n-1} \sin(nx)$  has upper bound  $\frac{2}{|1 - e^{i\epsilon}|}$ . ■

### Question 4: Convex functions, Arzelà–Ascoli theorem

Let  $(F_n : [-2, 2] \rightarrow \mathbb{R})$  be a uniformly bounded sequence of convex functions. Show that it has a subsequence that converges uniformly on  $[-1, 1]$ .

*Proof.* Because of **Arzelà–Ascoli theorem**, we only have to prove  $\{F_n|_{[-1,1]}\}$  is pointwise equicontinuous. Let  $M$  be a uniform bound of  $F_n$ . Fix  $x \in [-1, 1]$ . The proof follows from noting that if  $x < s$ , then

$$F(s) \leq \left( \frac{F(2) - F(x)}{2 - x} \right) (s - x) + F(x)$$

which implies

$$|F(s) - F(x)| \leq 2M |s - x|$$

and if  $s < x$ , then

$$F(s) \leq \left( \frac{F(x) - F(-2)}{x - (-2)} \right) (s - x) + F(x)$$

which implies

$$|F(s) - F(x)| \leq 2M |s - x|$$

■

### Question 5: inverse function theorem

Let  $U \subseteq \mathbb{R}^n$  be open, let  $C^1$  function  $f : U \rightarrow \mathbb{R}^n$  has globally full rank Jacobian, let  $V \subseteq \mathbb{R}^n$  be open, and let continuous  $g : V \rightarrow U$  satisfies  $f \circ g(x) = x$  for all  $x \in V$ . Show that  $g$  is also  $C^1$ .

*Proof.* Fix  $x \in V$ . By **inverse function theorem**, there exists some neighborhood  $B_\epsilon(g(x))$  of  $g(x)$  contained by  $U$  such that  $f$  maps  $B_\epsilon(g(x))$  injectively to an open set  $f(B_\epsilon(g(x)))$  with  $C^1$  inverse  $f^{-1} : f(B_\epsilon(g(x))) \rightarrow B_\epsilon(g(x))$ .

Because  $g$  is continuous, we know there exists some neighborhood  $B_\delta(x)$  of  $x$  contained by  $V$  such that  $g$  maps  $B_\delta(x)$  into  $B_\epsilon(g(x))$ . This give us

$$y = f \circ g(y) \in f(B_\epsilon(g(x))), \quad \text{for all } y \in B_\delta(x)$$

which implies  $B_\delta(x) \subseteq f(B_\epsilon(g(x)))$ . It remains to show  $g|_{B_\delta(x)} = f^{-1}|_{B_\delta(x)}$ , which follows from:

$$g(y) = f^{-1} \circ f \circ g(y) = f^{-1}(y), \quad \text{for all } y \in B_\delta(x)$$

■

## 2.9 Year 105

### Question 1

(i) Calculate

$$\sup_{x \neq 0 \in \mathbb{R}} \frac{x^2 - \ln(1 + x^2)}{x^4}$$

(ii) Calculate

$$\iint_A xy \sin(x^2 - y^2) dx dy$$

where  $A = \{(x, y) \in \mathbb{R}^2 : 0 < y < 1, y < x \text{ and } x^2 - y^2 < 1\}$ .

(iii) Prove that

$$\sum_{n=0}^{\infty} \frac{1}{(2n+1)^2} = \frac{\pi^2}{8}$$

*Proof.* (i): Noting the symmetry, we only have to analyze the continuous function  $f : \mathbb{R}^+ \rightarrow \mathbb{R}$  defined by

$$f(t) \triangleq \frac{t - \ln(1+t)}{t^2}$$

Using **L'Hospital Rule**, we see

$$\lim_{t \searrow 0} f(t) = \frac{1}{2}$$

Compute

$$f'(t) = \frac{t^2 - 2(1+t)(t - \ln(1+t))}{t^3(1+t)}$$

Consider the nominator  $h : \mathbb{R}^+ \rightarrow \mathbb{R}$  of derivative of  $f$ :

$$h(t) \triangleq t^2 - 2(1+t)(t - \ln(1+t)) = t^2 + 2(1+t)\ln(1+t) - 2t(1+t)$$

Clearly

$$h(0) = 0 \text{ and } h'(t) = 2\ln(1+t) - 2(1+t) + 2 \leq 0, \quad \text{for all } t > 0$$



We have shown  $f'(t) \leq 0$  for all  $t > 0$ . This implies  $\frac{1}{2}$  is indeed the supremum.

(ii): Note that

$$A = \left\{ (x, y) \in \mathbb{R}^2 : 0 < y < 1, y < x < \sqrt{1+y^2} \right\}$$

Therefore,

$$\begin{aligned} \iint_A xy \sin(x^2 - y^2) dx dy &= \int_0^1 \int_y^{\sqrt{1+y^2}} xy \sin(x^2 - y^2) dx dy \\ &= \int_0^1 \frac{1}{2} \int_0^1 y \sin(u) du dy, \quad (u = x^2 - y^2) \\ &= \frac{1}{2} (1 - \cos(1)) \int_0^1 y dy = \frac{1 - \cos(1)}{4} \end{aligned}$$

For (iii), recall that the solution of the **Basel's problem** is

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

Therefore,

$$\sum_{n=0}^{\infty} \frac{1}{(2n+1)^2} = \sum_{n=1}^{\infty} \frac{1}{n^2} - \sum_{n=1}^{\infty} \frac{1}{(2n)^2} = \frac{3}{4} \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{8}$$

■

## Question 2: Cantor set, standard

Let

$$C \triangleq \left\{ \sum_{n=1}^{\infty} \frac{a_n}{3^n} \in \mathbb{R} : a_n \in \{0, 2\} \right\}$$

Show that

- (i)  $C \subseteq [0, 1]$ .
- (ii)  $C$  has no interior point.
- (iii)  $C$  is compact.

*Proof.* (i) is clear. We now prove (ii). Fix  $\{a_n\}$ . If  $\{a_n\}$  have trailing 0, then we see for large  $N$ ,

$$\sum \frac{a_n}{3^n} + \frac{1}{3^N} + \frac{1}{3^{N+1}} \notin C$$

If  $\{a_n\}$  have trailing 2, then we see for large  $N$ ,

$$\sum \frac{a_n}{3^n} - \frac{1}{3^N} - \frac{1}{3^{N+1}} \notin C$$

Suppose  $\{a_n\}$  have no trailing number. We then see that  $(\sum \frac{a_n}{3^n}) + \frac{1}{3^N} \notin C$ .

We now prove (iii). Clearly  $C$  is bounded. Therefore by Hien-Borel, we only have to prove  $C$  is closed. Let  $x \in (0, 1) - C$ , and write

$$x = \sum_{n=1}^{\infty} \frac{b_n}{3^n}, \quad \text{where } b_n \in \{0, 1, 2\}$$

Let  $N$  be the smallest natural number that makes  $b_N = 1$ . Fix  $y = \sum \frac{a_n}{3^n} \in C$ . Suppose there is some  $n < N$  that makes  $a_n \neq b_n$ . Let  $n$  be the smallest natural number that makes  $a_n \neq b_n$ . If  $a_n = 0$ , then  $x - y \geq \frac{1}{3^N}$ . If  $a_n = 2$ , then  $y - x \geq \frac{1}{3^N}$ .

Suppose  $a_1 = b_1, a_2 = b_2, \dots, a_{N-1} = b_{N-1}$ . Clearly we can not have  $2 = b_{N+1} = b_{N+2} = \dots$  nor  $0 = b_{N+1} = b_{N+2} = \dots$ , otherwise  $x \in C$ . Let  $M_0 > N$  and  $M_2 > N$  be smallest that makes  $b_{M_0} = 0, b_{M_2} = 2$ . If  $a_N = 2$ , we see  $y - x \geq \frac{1}{3^{M_0}}$ . If  $a_N = 0$ , we see  $x - y \geq \frac{1}{3^{M_2}}$ . We have shown  $B_{\frac{1}{3^{\max\{M_0, M_2\}}}}(x) \notin C$ . ■

### Question 3

Let

$$A \triangleq \{f \in C^0[0, 1] : \|f\|_{C_0} \leq 1\}$$

and

$$B \triangleq \{f \in C^1[0, 1] : \|f\|_{C_1} \leq 1\}$$

where

$$\|f\|_{C^0} \triangleq \sup_{x \in [0, 1]} |f| \quad \text{and} \quad \|f\|_{C^1} \triangleq \sup_{x \in [0, 1]} |f| + \sup_{x \in [0, 1]} |f'(x)|$$

Show that

- (i)  $A$  is not compact.
- (ii) Every sequence in  $B$  has a convergent subsequence in  $A$ .

*Proof.* For (i), clearly the sequence  $\{x^n\}$  forms a counter example. (ii) is a consequence of **Arzelà–Ascoli theorem**. ■

#### Question 4

Let  $f : [0, 1] \rightarrow \mathbb{R}$  be continuous and define  $F : [0, 1] \rightarrow \mathbb{R}$  by

$$F(x) \triangleq \int_0^x f(t)dt$$

Suppose  $F(1) = 0$ . Show that the value

$$\sup_{g \in C^0[0,1]} \frac{\int_0^1 F(x)g(x)dx}{\left(\int_0^1 G^2(x)dx\right)^{\frac{1}{2}}}$$

exists, where  $G(x) \triangleq \int_0^x g(t)dt$ .

*Proof.* Because  $F(0) = F(1) = 0$ , integration by part give us

$$\int_0^1 Fg = - \int_0^1 fG = -\langle f, G \rangle$$

Therefore, by Cauchy-Schwarz, we have

$$\left| \frac{-\langle f, G \rangle}{\|G\|} \right| \leq \|f\|$$

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